

Online Appendix

Night and Day: Diurnal Warming and Structural Transformation in India

Vedarshi Shastry

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This appendix collects robustness exhibits for the diurnal temperature decomposition. The main text reports the baseline in surface-air minima and maxima (T_{min} , T_{max}); here the default measure is their wet-bulb counterpart (T_{wmin} , T_{wmax}) unless a table notes otherwise, so that the surface-air results in the main text and the wet-bulb results below can be read against one another. The tables sweep the climate measure, the averaging window and season, the fixed-effects and trend structure, and sample heterogeneity, and decompose worker shares into total, rural, and urban components.

Appendix A. Theoretical Framework

The empirical decomposition rests on a general-equilibrium framework that assigns nighttime warming ($T_{\min}$) to a land-biased shock and daytime warming ($T_{\max}$) to a Hicks-neutral productivity shock, embedded in a CES agricultural technology under two market closures. This appendix develops it and delimits what it can and cannot deliver.

A.1. Overview and Relation to the Empirics

The empirical decomposition isolates two sets of facts that any framework must confront.

Agricultural-side facts (ICRISAT districts). In the surface-air (dry-bulb) baseline, nighttime and daytime warming contract cultivated area by statistically indistinguishable amounts, but only daytime warming raises the local harvest price and depresses the field wage; cereal yields respond little under either margin. Under humidity-adjusted (wet-bulb) temperatures the two margins separate on the land side as well: nighttime warming raises yields and output and leaves area intact—the signature of land augmentation—while daytime warming sharply contracts both.

Labor-side facts (Census districts). The two margins carry *opposite* signs on the allocation of farm labor. A warmer night raises the agricultural wage-labor share and drains the cultivator and seasonal pools; a warmer day lowers the wage-labor share and refills the seasonal margin. The labor divergence is sharp in rural district populations and muted—approaching zero at longer horizons—in urban populations. In towns, over two-decade long differences, *both* margins depress non-agricultural employment.

This document formalizes a mechanism consistent with these facts and, equally importantly, delimits what the mechanism can and cannot deliver. The modeling choices are dictated by the empirics.

1. **CES, not Cobb–Douglas.** A land-biased shock and a neutral shock can move agricultural labor in opposite directions only if labor and land substitute with an elasticity $\sigma \neq 1$. Under Cobb–Douglas, a land-augmenting shock is observationally equivalent to a relabeled neutral shock (Lemma A2), and the day/night assignment has no implication for relative labor responses. We maintain $\sigma > 1$ (gross substitutes), the empirically relevant case for labor–land substitution through cultivation intensity.
2. **Land for $T_{\min}$, TFP for $T_{\max}$.** Nighttime warming acts on the land margin: it alters the productive services of soil and water on cultivated parcels and shifts the threshold at which marginal land is worth cultivating at all. Daytime extreme heat reduces total factor productivity and the physiological capacity for sustained field and outdoor work. This assignment is evaluated analytically in Section A.3.
3. **Two closures, one technology.** Rural and urban district aggregates share the ICRISAT production foundation but differ in their food-market closure. We derive a closed economy (local price clearing with Stone–Geary subsistence) and an open economy (district as a within-state price-taker), and bridge them with a single openness parameter $\chi \in [0, 1]$.

A central caution. A natural reading of the rural/urban contrast is that rural districts behave as closed economies and urban districts as open ones, and that this split *generates* the divergence in signs. Sections A.5–A.6 show this reading is not what a standard general-equilibrium model

delivers. The *sign* of the labor divergence is a factor-bias property that holds in both closures under the appropriate openness condition; the closed subsistence economy, far from producing the observed orientation, predicts its *reverse* (Proposition A4). The data—nights raise and days lower agricultural labor in both rural and urban populations—select the open regime in both. The closed/open distinction governs *amplitude* (Section A.9), not sign. Partial closure $0 < \chi < \chi^*$ does useful work nonetheless: it allows the harvest price to respond to output, which is why daytime—not nighttime—warming moves the price.

A.2. Environment

A region (a Census district, taken separately for rural and urban populations) produces an agricultural good a and a non-agricultural good n . The non-agricultural good is the numeraire ($p_n \equiv 1$); $p \equiv p_a$ is the relative price of food. Throughout, a hat denotes a log deviation from an initial equilibrium, $\hat{x} \equiv d \ln x$, and all comparative statics are first-order.

A.2.1. Agricultural technology

Agricultural output combines effective labor H_a and effective land through a CES aggregator scaled by total factor productivity:

$$(A1) \quad Y_a = A \left[\theta H_a^\rho + (1 - \theta) (BZ_c)^\rho \right]^{1/\rho}, \quad \rho \equiv \frac{\sigma - 1}{\sigma}, \quad \theta \in (0, 1).$$

Here $\sigma > 0$ is the elasticity of substitution between labor and land, Z_c is cultivated area (an extensive margin chosen by the farm; Section A.3.2), and BZ_c is *effective land*: B scales the productive services of a unit of cultivated area. The two temperature margins enter through distinct factors:

$$(A2) \quad A = A(T_{\max}), \quad B = B(T_{\min}),$$

with A a Hicks-neutral productivity shifter and B a land-augmentation index.

ASSUMPTION A1 (Technology and shocks).

- (i) Gross substitutes: $\sigma > 1$.
- (ii) Daytime heat lowers TFP: $A'(T_{\max}) < 0$.
- (iii) Nighttime warming is land-biased of ambiguous sign: $B'(T_{\min}) \geq 0$. As a benchmark we take $B'(T_{\min}) \geq 0$, the sign corroborated by the significantly positive humidity-adjusted (wet-bulb) night yield (Remark A5); the dry-bulb surface-air night response is on net contractionary and does not by itself identify the augmentation.

It is convenient to summarize the shocks by

$$(A3) \quad a \equiv \hat{A} = \frac{A'(T_{\max})}{A} dT_{\max} \quad (\leq 0 \text{ under warming}), \quad b \equiv \hat{B} = \frac{B'(T_{\min})}{B} dT_{\min} \quad (\geq 0; \geq 0 \text{ benchmark}).$$

Let s_L and $s_Z = 1 - s_L$ denote the revenue shares of labor and effective land at the initial equilibrium,

$$(A4) \quad s_L = \frac{\theta H_a^\rho}{\theta H_a^\rho + (1 - \theta) (BZ_c)^\rho}, \quad s_Z = 1 - s_L,$$

and define the composite

$$(A5) \quad \kappa \equiv \frac{\sigma}{s_Z} > 0.$$

A.2.2. Two roles for land

Nighttime warming acts on land in two conceptually distinct ways, which we keep separate. The *intensive* role is the augmentation index $B(T_{\min})$ in (A1): it raises or lowers the productive services of land already in cultivation, and hence influences yields. The *extensive* role is the viable cultivable extent: heat and moisture stress can push marginal parcels below the threshold at which cultivation breaks even. We write the cultivable endowment as $\bar{Z}(T_{\min}, T_{\max})$, non-increasing in both margins, and let the farm choose cultivated area $Z_c \leq \bar{Z}$ on a cost margin (Section A.3.2). This separation lets the augmentation and the extensive loss be distinguished by the humidity content of the heat. Under *wet-bulb* (humidity-adjusted) nighttime warming the augmentation channel is isolated: yields rise and cultivated area does *not* contract (Remark A5). Under *dry-bulb* surface-air warming the same nighttime margin is, on net, contractionary—area and output fall and yields are flat—because the dry-bulb measure loads the dry-heat moisture stress that wet-bulb nets out; it therefore does not identify the augmentation directly.

A.2.3. Non-agriculture

The non-agricultural sector produces the numeraire with decreasing returns to labor,

$$(A6) \quad Y_n = A_n L_n^\nu, \quad \nu \in (0, 1], \quad A_n = A_n(T_{\min}, T_{\max}),$$

where $\nu < 1$ captures congestion of a fixed non-labor factor and delivers a downward-sloping labor demand. Non-agricultural productivity falls with temperature through the physiological capacity for outdoor and physical informal work; we write $a_n \equiv \hat{A}_n \leq 0$ in both rural and urban areas. The difference across settings is one of observability, not of the shock: where agriculture is the dominant sector (rural districts) the within-agriculture reallocation swamps the small non-agricultural employment response, so a_n is second order and we set $a_n \approx 0$ for the rural analysis; where agriculture is a minor sector (towns, Section A.10) the same $a_n < 0$ becomes the primary observable margin.

A.2.4. Preferences and the food price

A representative household has Stone–Geary preferences over food and the numeraire with a subsistence floor $\bar{a} > 0$:

$$(A7) \quad U(c_a, c_n) = \alpha \ln(c_a - \bar{a}) + (1 - \alpha) \ln c_n.$$

Aggregating budget constraints in a closed regional economy, local market clearing pins the relative food price at

$$(A8) \quad p = \frac{\alpha}{1 - \alpha} \cdot \frac{Y_n}{Y_a - \bar{a} \bar{L}},$$

where $\bar{L}$ is the total workforce. Log-differentiating,

$$(A9) \quad \hat{p} = \hat{Y}_n - \mathcal{M} \hat{Y}_a, \quad \mathcal{M} \equiv \frac{Y_a}{Y_a - \bar{a}\bar{L}} > 1,$$

with $\mathcal{M}$ the *subsistence multiplier*: because households must eat, a contraction of food output raises its relative price more than one-for-one. We embed this in a general *openness* closure,

$$(A10) \quad \hat{p} = \chi(\hat{Y}_n - \mathcal{M} \hat{Y}_a), \quad \chi \in [0, 1],$$

where $\chi = 1$ is the closed economy (A9) and $\chi = 0$ is the open economy ($\hat{p} = 0$): the district takes the state-level food price as given because idiosyncratic supply shocks are arbitrated across districts within the state. Intermediate χ indexes partial integration.

A.2.5. Labor: the seasonal sink and the cultivator margin

The workforce $\bar{L}$ is allocated across four states whose population shares sum to one:

$$(A11) \quad l_a + l_c + l_n + l_s = 1,$$

where l_a is agricultural wage labor, l_c is self-cultivation, l_n is non-agriculture, and l_s is casual *seasonal* work. Field labor used in agricultural production is H_a , supplied by hired wage workers and family (cultivator) labor. The seasonal pool is the flexible residual margin.

Workers differ in a reservation value for seasonal or home work, with CDF $G(\cdot)$; an individual takes a formal wage job when the market wage w exceeds their reservation value. The reservation distribution applies to the *non-cultivating* population $\bar{L} - L_c$, so formal labor supply and the seasonal residual are

$$(A12) \quad L_a + L_n = (\bar{L} - L_c) G(w), \quad L_s = (\bar{L} - L_c) [1 - G(w)], \quad \eta_s \equiv \frac{w G'(w)}{G(w)} \geq 0,$$

with η_s the elasticity of market labor supply. The formal and seasonal states now sum to $\bar{L} - L_c$, leaving a positive mass $L_c > 0$ for self-cultivation and preserving the adding-up identity (A11); wage mobility equalizes the wage at w across the formal sectors.

The self-cultivation margin is pinned by a marginal-cultivator condition. A landholder works an own marginal parcel rather than hiring out only while the parcel's return $\rho(L_c)$ exceeds the market wage, with $\rho' < 0$ (parcels ordered by quality), so the marginal cultivator is indifferent, $\rho(L_c) = w$. A rise in the market wage raises the hiring-out cutoff and therefore *lowers* self-cultivation, $dL_c/dw < 0$ —the micro-foundation for the recomposition in Lemma A4. Until Section A.8, H_a denotes aggregate field labor and L_a its hired (agricultural-labourer) component, the census margin that clears the wage market; family labor on inframarginal own land passes the field-labor demand response through to the hired component, $\hat{L}_a = \hat{H}_a$. Because L_c moves second order relative to the formal-supply response, the market-clearing linearization (A21) is unchanged to first order, and the cultivator share adjusts through the composition identity of Section A.8 together with the marginal-cultivator condition.

A.3. The Producer's Problem

The representative farm chooses field labor H_a and cultivated area Z_c to maximize profit $pY_a - wH_a - c(Z_c)$, taking p and w as given, where $c(Z_c)$ is the cost of bringing land into cultivation,

$c' > 0, c'' \geq 0$.

A.3.1. The intensive margin: labor demand and factor bias

The first-order condition for field labor equates the value of its marginal product to the wage, $p \partial Y_a / \partial H_a = w$. The CES marginal product of labor is

$$(A13) \quad \text{MPL}_a = \frac{\partial Y_a}{\partial H_a} = \theta A^{(\sigma-1)/\sigma} \left(\frac{Y_a}{H_a} \right)^{1/\sigma}.$$

LEMMA A1 (Factor-bias decomposition). *To first order, holding cultivated area fixed,*

$$(A14) \quad \widehat{\text{MPL}}_a = a + \frac{s_Z}{\sigma} (b - \hat{H}_a),$$

and the farm's labor demand is

$$(A15) \quad \hat{H}_a = b + \kappa(\hat{p} + a - \hat{w}), \quad \kappa = \frac{\sigma}{s_Z}.$$

PROOF. From (A1), write $Y_a = A Q^{1/\rho}$ with $Q = \theta H_a^\rho + (1 - \theta)(B Z_c)^\rho$. Then $\partial Y_a / \partial H_a = A \theta Q^{1/\rho-1} H_a^{\rho-1}$. Using $1/\rho - 1 = 1/(\sigma - 1)$ and $Q^{1/(\sigma-1)} = (Y_a/A)^{1/\sigma}$ gives (A13). Taking logs: $\ln \text{MPL}_a = \ln \theta + \frac{\sigma-1}{\sigma} \ln A + \frac{1}{\sigma} (\ln Y_a - \ln H_a)$. Differentiating and using the production response at fixed area, $\hat{Y}_a = a + s_L \hat{H}_a + s_Z b$, gives $\hat{Y}_a - \hat{H}_a = a + s_Z (b - \hat{H}_a)$, hence $\widehat{\text{MPL}}_a = \frac{\sigma-1}{\sigma} a + \frac{1}{\sigma} [a + s_Z (b - \hat{H}_a)] = a + \frac{s_Z}{\sigma} (b - \hat{H}_a)$, which is (A14). The FOC in logs reads $\hat{p} + \widehat{\text{MPL}}_a = \hat{w}$; substituting (A14) and solving for $\hat{H}_a$ delivers (A15). $\square$

Equation (A15) is the engine of the paper. The land-bias shock b enters labor demand with a *direct unit coefficient*: a change in effective land quality shifts the marginal product of labor and hence the demand for labor at given prices, irrespective of whether the productivity change is positive or negative. The TFP shock a enters only through the “effective price” block $\kappa(\hat{p} + a - \hat{w})$: it is a pure scale force, raising or lowering the scale of operation without, on its own, altering the labor–land ratio. This is the precise sense in which *labor can adapt to $T_{\min}$ on its own terms* while $T_{\max}$ acts through scale. The distinction has bite only away from Cobb–Douglas:

LEMMA A2 (Cobb–Douglas is silent). *If $\sigma = 1$, the land-augmenting shock b is observationally equivalent to a Hicks-neutral shock scaled by $1 - \theta$. Labor demand depends only on the composite $\tilde{a} \equiv a + (1 - \theta)b$, and not on the composition of the shock between a and b . Consequently the day/night assignment carries no implication for the relative labor responses.*

PROOF. With $\sigma = 1$, $Y_a = A H_a^\theta (B Z_c)^{1-\theta} = [A B^{1-\theta}] H_a^\theta Z_c^{1-\theta}$, so the production function depends on a and b only through $\tilde{a} = a + (1 - \theta)b$. In (A15) with $\kappa = 1/s_Z = 1/(1 - \theta)$, $\hat{H}_a = b + \frac{1}{1-\theta}(\hat{p} + a - \hat{w}) = \frac{1}{1-\theta}(\hat{p} + \tilde{a} - \hat{w})$, a function of $\tilde{a}$ alone. $\square$

REMARK A1. *Lemma A2 is why the prior Cobb–Douglas formulation cannot, in principle, generate a day/night labor divergence from factor bias: under unit elasticity the two shocks are the same object up to scale. Assumption A1(i), $\sigma > 1$, is therefore load-bearing, not cosmetic. With $\sigma \neq 1$ the labor responses to a and b load on different structural elasticities (κ versus 1), and the composition of the shock matters.*

LEMMA A3 (Robustness to a labor-disaugmenting daytime shock). *Suppose daytime heat is modeled not as a Hicks-neutral loss but as labor-disaugmenting: effective field labor is $A_L(T_{\max}) H_a$ with*

$A'_L(T_{\max}) < 0$, so $a_L \equiv (A'_L/A_L) dT_{\max} \leq 0$ presses directly on the labor input, matching the physiological reading of heat stress. Then the labor-demand schedule becomes

$$(A16) \quad \hat{H}_a = b + \kappa(\hat{p} - \hat{w}) + \frac{\sigma - 1 + s_L}{s_Z} a_L,$$

in which the daytime coefficient $(\sigma - 1 + s_L)/s_Z > 0$ under Assumption A1(i), so $a_L < 0$ lowers agricultural labor demand—the same sign as the Hicks-neutral case—while the night term b still enters with a unit coefficient. The night/day divergence of Proposition A2 therefore does not depend on the daytime assignment.

PROOF. With $Y_a = A[\theta(A_L H_a)^\rho + (1 - \theta)(BZ_c)^\rho]^{1/\rho}$, the marginal product of labor is $MPL_a = \theta A^{(\sigma-1)/\sigma} A_L^\rho (Y_a/H_a)^{1/\sigma}$, so to first order $\widehat{MPL}_a = \frac{\sigma-1}{\sigma} a_L + \frac{1}{\sigma} (\hat{Y}_a - \hat{H}_a)$ (here $a = 0$, isolating the labor-bias channel). At fixed area $\hat{Y}_a = s_L(a_L + \hat{H}_a) + s_Z b$, so $\hat{Y}_a - \hat{H}_a = s_L a_L + s_Z b - s_Z \hat{H}_a$. Substituting into the log FOC $\hat{p} + \widehat{MPL}_a = \hat{w}$ and using $\kappa = \sigma/s_Z$ delivers (A16). Since $\sigma > 1$ and $s_L \in (0, 1)$, the coefficient $\sigma - 1 + s_L > 0$. $\square$

REMARK A2. The two daytime specifications differ only in the positive scalar weight on the day shock (κ under Hicks-neutrality, $(\sigma - 1 + s_L)/s_Z$ under labor-disaugmentation), so every sign prediction of the open-economy solution is preserved. We retain the Hicks-neutral form as the baseline because daytime heat also depresses non-labor productivity—crop metabolic stress and moisture loss during the hottest hours—so it is a deliberately conservative reduced form rather than a claim that heat spares the other factors. The physiological narrative and the formal assignment are thus consistent up to this innocuous reparameterization.

A.3.2. The extensive margin: cultivated area

The first-order condition for cultivated area equates the value of the marginal product of physical land to its marginal cultivation cost, $p \partial Y_a / \partial Z_c = c'(Z_c)$. Writing the elasticity of marginal cost as $\zeta \equiv d \ln c' / d \ln Z_c \geq 0$, and including an upward shift in marginal cost of size $\mu_T dT \geq 0$ for each margin $T \in \{T_{\min}, T_{\max}\}$ (reflecting declining viable extent), the extensive margin satisfies

$$(A17) \quad \hat{p} + a + b + \frac{s_L}{\sigma} (\hat{H}_a - b - \hat{Z}_c) = \zeta \hat{Z}_c + \mu_T dT.$$

PROPOSITION A1 (Common area contraction). Both temperature margins contract cultivated area. At the equilibrium response of $\hat{H}_a$,

$$(A18) \quad \frac{\partial \ln Z_c}{\partial T_{\min}} \approx \frac{\partial \ln Z_c}{\partial T_{\max}} < 0.$$

The two semi-elasticities can be made equal by appropriate calibration of $(\mu_{T_{\min}}, \mu_{T_{\max}})$, consistent with the data's failure to reject equality of the area coefficients.

PROOF. Solve (A17) for $\hat{Z}_c$ with the cost-shifter:

$$\left(\zeta + \frac{s_L}{\sigma}\right) \hat{Z}_c = \hat{p} + a + b \left(1 - \frac{s_L}{\sigma}\right) + \frac{s_L}{\sigma} \hat{H}_a - \mu_T dT.$$

Under $T_{\max}$: the direct TFP term $a < 0$ and the cost shifter $-\mu_{T_{\max}} dT_{\max} < 0$ both reduce the right-hand side. The endogenous price response $\hat{p} > 0$ pushes in the opposite direction, but in the empirically selected partial-open regime ($\chi < \chi^*$, Proposition A5) it is bounded: $\hat{p} = \chi(\hat{Y}_n - \mathcal{M}\hat{Y}_a)$

is proportional to the small openness weight χ , so $|\hat{p}|$ is dominated by the direct loss $|a|$ and the cost shifter for χ below the threshold—the same range the observed labor signs require. Hence the right-hand side falls. Under $T_{\min}$: the cost shifter $-\mu_{T_{\min}} dT_{\min} < 0$ dominates the augmentation term $b(1 - s_L/\sigma)$ —with $\hat{p} \approx 0$ at the nighttime margin—whenever $|b|$ is small or $\mu_{T_{\min}}$ is sufficiently large. Hence $\hat{Z}_c < 0$ for both margins. Equality of the two semi-elasticities is a calibration of $(\mu_{T_{\min}}, \mu_{T_{\max}})$ against the direct TFP term, and is supported by the area F -test in Table 1. $\square$

REMARK A3 (Area is common; labor is divergent). *Proposition A1 delivers the first stylized fact—area falls equally under both margins—through the extensive channel shared by both shocks. The day/night divergence lives entirely in the intensive reallocation of labor. Because the Census labor share reflects labor composition and intensity on cultivated land rather than total field hours, it is governed by (A15) and is not mechanically dragged down by area contraction: labor intensity $\hat{H}_a - \hat{Z}_c = b + \kappa(\hat{p} + a - \hat{w})$ is independent of the extensive margin. We therefore analyze labor reallocation at fixed area and treat the area result as the separate common margin.*

A.4. Closing the Economy

Collecting the equilibrium conditions in log-deviations (area fixed for the intensive labor block, per Remark A3):

$$(A19) \quad (\text{agricultural labor demand}) \quad \hat{L}_a = b + \kappa(\hat{p} + a - \hat{w}),$$

$$(A20) \quad (\text{non-agricultural labor demand}) \quad \hat{L}_n = \frac{a_n - \hat{w}}{1 - \nu},$$

$$(A21) \quad (\text{market labor supply}) \quad \phi_a \hat{L}_a + \phi_n \hat{L}_n = \eta_s \hat{w},$$

$$(A22) \quad (\text{food price}) \quad \hat{p} = \chi(\hat{Y}_n - \mathcal{M} \hat{Y}_a),$$

$$(A23) \quad (\text{outputs}) \quad \hat{Y}_a = a + s_L \hat{L}_a + s_Z b, \quad \hat{Y}_n = \nu \hat{L}_n,$$

where $\phi_a = L_a/(L_a + L_n)$ and $\phi_n = 1 - \phi_a$ are shares of market labor, and (A20) follows from the non-agricultural first-order condition $w = \nu A_n L_n^{\nu-1}$. The unknowns are $\{\hat{L}_a, \hat{L}_n, \hat{w}, \hat{p}\}$. For the district analysis we set $a_n = 0$ and define

$$(A24) \quad \Omega \equiv \eta_s + \frac{\phi_n}{1 - \nu} > 0.$$

A.5. The Open Economy

Set $\chi = 0$, so $\hat{p} = 0$. Substituting (A19)–(A20) into (A21) and solving,

$$(A25) \quad \hat{w} = \frac{\phi_a(\kappa a + b)}{\Delta_o}, \quad \hat{L}_a = \frac{\Delta'_o}{\Delta_o}(\kappa a + b),$$

where $\Delta_o \equiv \eta_s + \phi_a \kappa + \phi_n/(1 - \nu) > 0$ and $\Delta'_o \equiv \Delta_o - \phi_a \kappa = \eta_s + \phi_n/(1 - \nu) > 0$, so $0 < \Delta'_o/\Delta_o < 1$.

PROPOSITION A2 (Day/night divergence in the open economy). *Under Assumption A1, agricultural*

labor rises with nighttime warming and falls with daytime warming:

$$(A26) \quad \frac{\partial \hat{L}_a}{\partial T_{\min}} = \frac{\Delta'_o}{\Delta_o} \cdot \frac{B'(T_{\min})}{B} \geq 0, \quad \frac{\partial \hat{L}_a}{\partial T_{\max}} = \frac{\Delta'_o}{\Delta_o} \cdot \kappa \cdot \frac{A'(T_{\max})}{A} < 0,$$

so the two derivatives have opposite signs whenever $B'(T_{\min}) > 0$ and $A'(T_{\max}) < 0$. The seasonal margin moves oppositely to agricultural labor; the market wage shares the sign of $\kappa a + b$; and non-agricultural labor moves with $-\hat{w}$.

PROOF. Differentiate (A25). Since $a = (A'/A)dT_{\max}$ and $b = (B'/B)dT_{\min}$, the derivatives follow with the stated signs given $A' < 0$, $B' \geq 0$, and $\Delta_o, \Delta'_o, \kappa > 0$. From (A25), $\hat{w}$ has the sign of $\kappa a + b$; from (A20) with $a_n = 0$, $\hat{L}_n = -\hat{w}/(1 - \nu)$ has the sign of $-(\kappa a + b)$; the seasonal residual $\hat{L}_s = -[(L_a + L_n)\eta_s \hat{w}]/L_s$ shares the sign of $-\hat{w}$, hence opposes $\hat{L}_a$. $\square$

REMARK A4. The opposite signs arise because b enters (A19) directly (factor-bias channel) while a enters through the scale block $\kappa(\cdot)$. Lemma A2 shows this asymmetry requires $\sigma \neq 1$: for $\sigma \neq 1$ the responses load on 1 versus κ , which are distinct for any $s_Z \neq 1$.

PROPOSITION A3 (Labor's own condition). The sign of the agricultural-labor response to nighttime warming is governed by the land-bias of the shock and the labor-market closure, not by the magnitude of the yield response. Specifically:

- (i) $\text{sgn}(\partial \hat{L}_a / \partial T_{\min}) = \text{sgn}(B'(T_{\min}))$, with magnitude Δ'_o / Δ_o determined by $(\sigma, \eta_s, \nu, \text{shares})$;
- (ii) the yield response $\widehat{\text{yield}} = a + s_L(\hat{L}_a - \hat{Z}_c) + s_Z b$ is weakly positive under nighttime warming when the augmentation is isolated ($b \geq 0$, $\hat{L}_a > 0$), so the model predicts a positive night yield; this is borne out under humidity-adjusted (wet-bulb) temperatures, where the night yield is significantly positive and area does not contract. Under the dry-bulb measure the night yield is instead flat and area and output fall, because the surface-air margin loads a dry-heat moisture stress; the prediction is thus confirmed in wet-bulb rather than read off the dry-bulb estimate (Remark A5);
- (iii) under daytime warming the labor and output responses necessarily share a sign (pure scale): $\text{sgn}(\partial \hat{L}_a / \partial T_{\max}) = \text{sgn}(\partial \hat{Y}_a / \partial T_{\max}) = \text{sgn}(A'(T_{\max})) < 0$.

PROOF. Parts (i) and (iii) are immediate from Proposition A2 and $\hat{Y}_a = a + s_L \hat{L}_a + s_Z b$ (with $a < 0$, $\hat{L}_a < 0$ under $T_{\max}$, both terms negative). For (ii), substitute $\hat{L}_a = (\Delta'_o / \Delta_o)b$ (nighttime, $a = 0$) into the yield expression: $\widehat{\text{yield}} = s_L[(\Delta'_o / \Delta_o)b - \hat{Z}_c] + s_Z b > 0$, since $\Delta'_o / \Delta_o > 0$, $b \geq 0$, and $\hat{Z}_c < 0$ (Proposition A1): the labor and land-bias terms are strictly positive and the area term adds to them. The model thus predicts a strictly positive night yield, whose magnitude is smaller than and distinct from the labor response (it nets the bias against scale and area). This prediction is borne out under wet-bulb, where the night yield is significantly positive; the dry-bulb estimate, which loads the dry-heat extensive loss, instead reads flat-to-contractionary (Remark A5). $\square$

Proposition A3 formalizes the verbal claim that “ $T_{\min}$ can raise or lower land productivity, but labor has its own condition for whether agricultural labor rises.” Daytime warming admits no such separation: it is neutral, so labor must follow scale.

REMARK A5 (Evidence on the sign of B'). Within the model the nighttime labor response is a sign test for the land bias ($\text{sgn}(\partial \hat{L}_a / \partial T_{\min}) = \text{sgn}(B'(T_{\min}))$), and this divergence is cleanly identified: in the surface-air (dry-bulb) baseline the two margins are near-orthogonal after fixed effects (within-transformed VIF ≈ 1), so the labor signs are not a collinearity artifact. The direct evidence on $B' > 0$ comes from the humidity-adjusted (wet-bulb) night yield, which is significantly positive (+0.15) and is not accompanied

by an area contraction—wet-bulb isolates the augmentation. The surface-air night margin does not supply this evidence: there nighttime warming contracts area and output and leaves yield flat, because the dry-bulb measure loads a dry-heat moisture stress (high evaporative demand) that the wet-bulb measure nets out. We therefore do not read the flat dry-bulb yield as confirming augmentation; the augmentation is a wet-bulb finding, and because the wet-bulb pair is far more collinear than the dry-bulb baseline ($\rho \approx 0.97$, $\text{VIF} \approx 3.4$, versus $\text{VIF} \approx 1$), it is corroborative rather than a second clean identification. The model accordingly guides the cleanly-identified labor signs; the land-augmentation channel behind them is supported but not structurally estimated. A deeper reading—that the land bias is the water-stress (evaporative-demand) component of warming, which the dry-versus-wet-bulb contrast isolates—is left to separate work.

COROLLARY A1 (Open-economy prices and wages). *At $\chi = 0$ the harvest price is fixed and the field wage moves as in Proposition A2: $\partial\hat{w}/\partial T_{\max} < 0$ (TFP loss collapses the marginal product of labor) and $\partial\hat{w}/\partial T_{\min} \geq 0$ but small (the land bias raises labor demand modestly through the wage). A non-zero harvest-price response requires partial closure, $\chi > 0$.*

A.6. The Closed Economy and the Openness Threshold

Now let $\hat{p}$ be endogenous. Substituting (A22) with $\hat{Y}_n = -v\hat{w}/(1-v)$ and $\hat{Y}_a = a + s_L\hat{L}_a + s_Zb$ into (A19), using $\hat{w} = \phi_a\hat{L}_a/\Omega$ from (A21), and collecting terms (full algebra in Appendix A.13) yields the general solution

$$(A27) \quad \hat{L}_a = \frac{1 - \sigma\chi\mathcal{M}}{D(\chi)} b + \frac{\kappa(1 - \chi\mathcal{M})}{D(\chi)} a,$$

where $D(\chi) \equiv 1 + \kappa\chi\mathcal{M}s_L + \kappa\zeta(\chi)\phi_a/\Omega > 0$, $\zeta(\chi) \equiv 1 + \chi v/(1-v)$, and we used $\kappa s_Z = \sigma$. Setting $\chi = 0$ recovers (A25); setting $\chi = 1$ gives the fully closed economy.

PROPOSITION A4 (Closed economy reverses the orientation). *In the fully closed economy ($\chi = 1$) with subsistence ($\mathcal{M} > 1$) and $\sigma > 1$,*

$$(A28) \quad \frac{\partial\hat{L}_a}{\partial T_{\min}} < 0 \quad \text{and} \quad \frac{\partial\hat{L}_a}{\partial T_{\max}} > 0:$$

nighttime warming lowers and daytime warming raises agricultural labor—the reverse of the open economy and of the data.

PROOF. At $\chi = 1$, from (A27) the coefficient on a is $\kappa(1 - \mathcal{M})/D(1) < 0$ (since $\mathcal{M} > 1$), so $\partial\hat{L}_a/\partial T_{\max} = [\kappa(1 - \mathcal{M})/D(1)](A'/A) > 0$ as $A' < 0$; the coefficient on b is $(1 - \sigma\mathcal{M})/D(1) < 0$ (since $\sigma\mathcal{M} > \sigma > 1$), so $\partial\hat{L}_a/\partial T_{\min} = [(1 - \sigma\mathcal{M})/D(1)](B'/B) \leq 0$ as $B' \geq 0$. Both orientations reverse. The mechanism is the subsistence multiplier: a TFP-driven output contraction raises the food price more than one-for-one, generating an income effect that pulls labor *into* agriculture rather than out. $\square$

Proposition A4 is the model's central caution. A closed subsistence economy does not amplify the observed divergence; it inverts it. Comparing the two closures pins down exactly how open the economy must be to deliver the data-consistent orientation.

PROPOSITION A5 (Openness threshold). *The divergence has the observed orientation—nighttime warming raising and daytime warming lowering agricultural labor (strictly when $B' > 0$)—if and only if*

$$(A29) \quad \chi < \chi^* \equiv \frac{1}{\sigma\mathcal{M}}.$$

Because $\sigma > 1$, the binding condition is the nighttime sign: the daytime sign is correct for the wider range $\chi < 1/\mathcal{M}$, while the nighttime sign requires the tighter $\chi < 1/(\sigma\mathcal{M})$.

PROOF. From (A27), $\partial\hat{L}_a/\partial T_{\max}$ has the sign of $\kappa(1 - \chi\mathcal{M})(A'/A)$, which is negative iff $1 - \chi\mathcal{M} > 0$, i.e. $\chi < 1/\mathcal{M}$. And $\partial\hat{L}_a/\partial T_{\min}$ has the sign of $(1 - \sigma\chi\mathcal{M})(B'/B)$, which is ≥ 0 (for $B' \geq 0$) iff $1 - \sigma\chi\mathcal{M} \geq 0$, i.e. $\chi \leq 1/(\sigma\mathcal{M})$. The intersection is $\chi < \chi^* = 1/(\sigma\mathcal{M})$; since $\sigma > 1$, $\chi^* < 1/\mathcal{M}$, so the nighttime condition binds. $\square$

REMARK A6 (The data select the open regime; closed/open governs amplitude). Empirically, nighttime warming raises and daytime warming lowers agricultural labor in both rural and urban populations (Section A.1). By Proposition A5 this requires $\chi < \chi^*$ in both—both populations are sufficiently integrated into the within-state food market. The closed economy is rejected in both. Consequently the rural/urban difference is not a sign difference; it is an amplitude difference (Section A.9). Partial closure $0 < \chi < \chi^*$ nonetheless does useful work: it allows the harvest price to respond to idiosyncratic output shocks, which is why daytime—not nighttime—warming moves the price.

A.7. Comparative Statics for the Agricultural Margins

We record the agricultural-economy outcomes (area, yield, field wage, harvest price) in the empirically selected regime $0 < \chi < \chi^*$.

Area. By Proposition A1, $\partial \ln Z_c / \partial T_{\min} \approx \partial \ln Z_c / \partial T_{\max} < 0$: a common contraction under both margins.

Yield. With $\widehat{\text{yield}} = a + s_L(\hat{L}_a - \hat{Z}_c) + s_Z b$: under $T_{\min}$ ($a = 0, b \geq 0$, intensity up) yield is weakly positive; under $T_{\max}$ ($a < 0$ dominant) yield is weakly negative. In dry-bulb these read small and imprecise—the night augmentation offset by the dry-heat extensive loss—whereas under wet-bulb, which strips that loss, the predicted positive night yield is realized (+0.15, significant).

Field wage. The market wage is $\hat{w} = \phi_a \hat{L}_a / \Omega$, sharing the sign of $\hat{L}_a$. Hence $\partial \hat{w} / \partial T_{\max} < 0$ (large: the neutral TFP loss collapses the marginal product of labor) and $\partial \hat{w} / \partial T_{\min} \geq 0$ but small. This reproduces the significant negative daytime wage effect and the flat nighttime wage.

Harvest price. From (A22), $\hat{p} = \chi(\hat{Y}_n - \mathcal{M}\hat{Y}_a)$. The price rises when food output contracts. The daytime contraction is larger and operates through productivity (yields fall), so $\partial \hat{p} / \partial T_{\max} > 0$ and sizable; the nighttime contraction is buffered by intensification (yields hold up), so $\partial \hat{p} / \partial T_{\min}$ is approximately zero and empirically indistinguishable from it. The price channel thus differentiates the two margins, consistent with the data's rejection of price coefficient equality.

PROPOSITION A6 (The four agricultural margins). In the regime $0 < \chi < \chi^*$ under Assumption A1:

	Area	Yield	Field wage	Harvest price
$T_{\min}$	(−)	($\approx 0, +$)	(≈ 0)	(≈ 0)
$T_{\max}$	(−)	($\approx 0, −$)	(−)	(+)

with area common across margins and the wage–price wedge opening only under daytime warming.

PROOF. Collecting the four paragraphs above: area from Proposition A1; yield from the displayed decomposition; wage from $\hat{w} = \phi_a \hat{L}_a / \Omega$ and Proposition A2; price from (A22) with the larger, productivity-driven daytime output contraction. $\square$

A.8. Within-Agriculture Composition

The Census distinguishes self-cultivators (l_c) from agricultural wage laborers (l_a), and both from seasonal workers (l_s). Landowning households supply family labor to their own inframarginal land and decide, at the margin, whether to hire out at the wage w ; the marginal cultivator is indifferent between working an own marginal parcel and earning w . Hired wage labor L_a is the demand margin that clears the wage market; seasonal work is the residual buffer (A12).

LEMMA A4 (Recomposition toward wage labor). *Let $dl_n \approx 0$ (district non-agriculture flat in the panel). Then by the adding-up identity (A11),*

$$(A30) \quad dl_c + dl_s = -dl_a = -l_a \hat{L}_a.$$

Under nighttime warming $\hat{L}_a > 0$, so $dl_c + dl_s < 0$: the rise in wage labor is supplied by a fall in self-cultivation (marginal cultivators hire out) and a drain of the seasonal pool. Under daytime warming $\hat{L}_a < 0$, so $dl_c + dl_s > 0$: the shed wage labor accumulates in the seasonal pool, with the seasonal margin—being the more elastic buffer (η_s large)—absorbing the bulk, and self-cultivation moving little.

PROOF. Differentiate (A11) with $dl_n = 0$ to get $dl_a + dl_c + dl_s = 0$. The sign follows from Proposition A2. The split between dl_c and dl_s is governed by the relative elasticities: when seasonal supply is highly elastic it clears most of the adjustment, so $|dl_s| \gg |dl_c|$, matching the larger seasonal coefficient in the data. $\square$

Lemma A4 reproduces the composition pattern of Table 1, Panel B: nighttime warming raises wage labor while lowering both cultivation and seasonal shares (the “commercializing” farm that retains and reorganizes labor), and daytime warming reverses the wage-labor flow back onto the seasonal margin.

A.9. Rural versus Urban: Amplitude

Both rural and urban district aggregates satisfy $\chi < \chi^*$ (Remark A6), so the *sign* of every response is common. They differ in the *magnitude* of the agricultural-labor-share response $dl_a = l_a \hat{L}_a$ through two channels.

PROPOSITION A7 (Rural amplification). *The agricultural-labor-share response is larger in absolute value in rural than in urban populations, with the same sign:*

$$(A31) \quad |dl_a^{rural}| > |dl_a^{urban}|,$$

driven by:

- (i) *The agricultural employment share l_a , which is several times larger in rural areas, scales the elasticity $\hat{L}_a$ into a larger share change; and*
- (ii) *The seasonal-supply elasticity η_s , which is larger in rural areas where the casual seasonal margin is deep. Since $\partial(\Delta'_o/\Delta_o)/\partial\eta_s > 0$, a larger rural η_s raises the elasticity $\hat{L}_a$ toward its upper bound of one.*

Both channels enlarge $|dl_a|$ rurally while preserving the sign.

PROOF. From Proposition A2, $\hat{L}_a = (\Delta'_o/\Delta_o)(\kappa a + b)$ and $dl_a = l_a \hat{L}_a$. Channel (i): dl_a scales linearly in l_a . Channel (ii): $\Delta'_o/\Delta_o = (\eta_s + \phi_n/(1 - \nu))/(\eta_s + \phi_a\kappa + \phi_n/(1 - \nu))$ increases in η_s toward 1. Both enlarge $|dl_a|$ in rural areas. $\square$

The rural/urban gap is an amplitude gap built on a common factor-bias sign—the pattern in the data, where the divergence F -statistic is large for rural workers and small for urban workers, with the same sign structure in both.

A.10. Towns and the Non-Agricultural Drag

Towns are open economies in which agriculture is a small share of employment, so the agricultural-labor divergence is attenuated to insignificance through Channel (i) of Proposition A7. The salient town outcome is instead on the non-agricultural side, where *both* margins depress employment over long differences. This follows once daytime—and, through sleep disruption and next-day heat strain, nighttime—warming reduces the physiological capacity for outdoor and physical informal work that dominates the lower end of urban services.

PROPOSITION A8 (Non-agricultural drag dominates in towns). *Let non-agricultural productivity fall under both margins, $a_n = \hat{A}_n < 0$ for both $T_{\min}$ and $T_{\max}$. Then in the open town economy non-agricultural employment falls under both margins,*

$$(A32) \quad \frac{\partial \hat{L}_n}{\partial T_{\min}} < 0 \quad \text{and} \quad \frac{\partial \hat{L}_n}{\partial T_{\max}} < 0,$$

so the common drag is the dominant term and the day/night divergence in non-agriculture is attenuated.

PROOF. Retaining $a_n < 0$ and re-solving (A19)–(A21) at $\chi \approx 0$ (Appendix A.13, equation (A33)): $\hat{L}_n = (a_n - \hat{w})/(1 - \nu)$ with $\hat{w} = [\phi_a(\kappa a + b) + \phi_n a_n/(1 - \nu)]/\Delta_o$. The direct effect of $a_n < 0$ on $\hat{L}_n$ is

$$\frac{1}{1 - \nu} \left(1 - \frac{\phi_n/(1 - \nu)}{\Delta_o} \right) a_n = \frac{\eta_s + \phi_a \kappa}{(1 - \nu)\Delta_o} a_n < 0,$$

which holds for both margins since $a_n < 0$ enters identically; the agricultural terms $(\kappa a + b)$ act only through $\hat{w}$ and cannot reverse the sign. Hence $\partial \hat{L}_n / \partial T < 0$ for both $T \in \{T_{\min}, T_{\max}\}$. $\square$

The town result is therefore not a failure of the day/night mechanism but a different margin: where agriculture is too small to register the reallocation, the common heat-capacity loss in informal urban work produces a non-agricultural contraction under *both* margins as the dominant term—a drag on structural transformation shared by both ends of the diurnal cycle—while any residual day/night divergence is second order relative to the agricultural reallocation and need not vanish exactly (it resurfaces, for instance, in the wet-bulb and Indo-Gangetic-North cuts).

REMARK A7 (An unmodeled town pattern). *Proposition A8 predicts a non-agricultural contraction under both margins, with the agricultural divergence attenuated to near-zero in towns, because town agriculture is too small to register the factor-bias reallocation. The peninsular-south subsample departs from this: nighttime warming raises the town agricultural wage-labor share sharply in the regional town long difference. The log-linear framework does not generate this pull onto the land at the town scale; it is a robust empirical pattern that the model brackets rather than explains, and the paper treats it as suggestive rather than as a test of the mechanism.*

A.11. What Is Load-Bearing

Table A1 records how each result depends on the primitives, isolating the assumptions that do the work and the knife-edges that should be stated clearly.

Result	Depends on	Comment
Factor-bias content (Lm. A1, A2)	$\sigma \neq 1$	Cobb–Douglas makes b a relabeled neutral shock; $\sigma > 1$ is load-bearing.
Open divergence (Pr. A2)	$A' < 0, B' \geq 0, \chi < \chi^*$	Sign needs only the channel assignment and openness; holds for any $\sigma > 0$.
Labor's own condition (Pr. A3)	land-bias of $T_{\min}$	Labor sign tracks B' and the closure, decoupled from yield magnitude.
Common area (Pr. A1)	extensive shifters μ_T	Equality of the two area effects is calibration, not a theorem.
Closed reversal (Pr. A4)	$\mathcal{M} > 1$	Subsistence inverts the orientation; the central caution.
Threshold $\chi^* = 1/(\sigma\mathcal{M})$ (Pr. A5)	$\sigma, \mathcal{M}$	$\sigma > 1$ makes the nighttime sign the binding condition.
Rural amplitude (Pr. A7)	l_a, η_s	Employment share and seasonal depth, not the food-price closure.
Town drag (Pr. A8)	$a_n < 0$ both margins	Requires a heat-capacity channel in informal non-agricultural work.

TABLE A1. Sensitivity of each result to the primitives.

A.12. Mapping to the Estimates

Table A2 links each model prediction to the corresponding reduced-form coefficient. The framework is a guide to *signs* and to the *ordering* of magnitudes (rural $>$ urban; daytime price–wage wedge $>$ nighttime); it is not a structural model and does not pin down levels. Several quantities held fixed here—nonlinear temperature thresholds, spatial sorting, migration—move discontinuously in the data and lie beyond a log-linear framework.

A.13. Closed- and Open-Economy Algebra

Derivation of the general solution (A27). Substitute $\hat{Y}_n = -v\hat{w}/(1-v)$ and $\hat{Y}_a = a + s_L\hat{L}_a + s_Zb$ into (A22):

$$\hat{p} = \chi \left[-\frac{v\hat{w}}{1-v} - \mathcal{M}(a + s_L\hat{L}_a + s_Zb) \right].$$

Insert into (A19):

$$\hat{L}_a = b + \kappa \left(\chi \left[-\frac{v\hat{w}}{1-v} - \mathcal{M}a - \mathcal{M}s_L\hat{L}_a - \mathcal{M}s_Zb \right] + a - \hat{w} \right).$$

Define $\xi(\chi) \equiv 1 + \chi v/(1-v)$. Collecting $\hat{L}_a$ and the $\hat{w}$ terms:

$$\hat{L}_a(1 + \kappa\chi\mathcal{M}s_L) = (1 - \kappa\chi\mathcal{M}s_Z)b + \kappa(1 - \chi\mathcal{M})a - \kappa\xi(\chi)\hat{w}.$$

From (A21) with $a_n = 0$, $\phi_a\hat{L}_a = \Omega\hat{w}$, so $\hat{w} = \phi_a\hat{L}_a/\Omega$. Substituting and collecting:

$$\hat{L}_a D(\chi) = (1 - \sigma\chi\mathcal{M})b + \kappa(1 - \chi\mathcal{M})a,$$

where $D(\chi) \equiv 1 + \kappa\chi\mathcal{M}s_L + \kappa\xi(\chi)\phi_a/\Omega > 0$, using $\kappa s_Z = \sigma$. This is (A27).

Prediction	Result	Empirical counterpart
Area contracts under both margins	Pr. A1	$T_{\min}, T_{\max}$ area coefficients (equality is calibrated, not predicted; $F = 0.01$).
Only daytime warming raises price, lowers wage (<i>over-identifying test</i>)	Pr. A6	Harvest price and field wage panels; price equality rejected ($F = 13.34$).
Nights raise, days lower agricultural wage labor	Pr. A2	Ag. labor share; divergence rejected ($F = 24.91$).
Positive night yield, isolated under wet-bulb	Pr. A3, Rem. A5	Wet-bulb night yield +0.15 (sig., area not contracting); dry-bulb night net-contractionary.
Wage labor drawn from cultivation & seasonal (nights); shed to seasonal (days)	Lm. A4	Cultivation and seasonal share panels (Table 1, Panel B).
Rural amplifies, urban mutes; same sign structure	Pr. A7	Rural vs. urban worker panels; $F = 26.34$ vs. $F = 6.05$ (ag. labor).
Closed economy would reverse signs	Pr. A4	Not observed $\Rightarrow$ districts are open ($\chi < \chi^*$).
Towns: both margins depress non-agricultural employment	Pr. A8	Town long-difference non-ag. share ($-0.038^{**}, -0.078^{***}$).

TABLE A2. Model predictions and their empirical counterparts.

Checking limits. At $\chi = 0$: $D(0) = 1 + \kappa\phi_a/\Omega$, $(1 - \sigma \cdot 0 \cdot \mathcal{M})b = b$, $\kappa(1 - 0 \cdot \mathcal{M})a = \kappa a$. So $\hat{L}_a = (b + \kappa a)/(1 + \kappa\phi_a/\Omega) = (\Delta'_o/\Delta_o)(\kappa a + b)$, recovering (A25).

Open economy with non-agricultural shock. Retaining $a_n \neq 0$ in (A20) and re-solving (A19)–(A21) at $\chi = 0$:

$$\begin{aligned}
(A33) \quad \hat{w} &= \frac{\phi_a(\kappa a + b) + \phi_n a_n / (1 - \nu)}{\Delta_o}, \\
\hat{L}_a &= \frac{\Delta'_o}{\Delta_o}(\kappa a + b) - \frac{\kappa\phi_n}{(1 - \nu)\Delta_o} a_n, \\
\hat{L}_n &= \frac{a_n - \hat{w}}{1 - \nu}.
\end{aligned}$$

These are used in Proposition A8 with $a_n < 0$ for both margins.

Appendix B. Empirical Strategy

This section derives the panel estimator used throughout the paper, the two standard-error families it is paired with (district clustering and Conley spatial-temporal HAC), and the simulation that selects the inference procedure by its empirical *size*. We also explain why the estimator’s state-specific linear trends are economically, not merely statistically, motivated.

Caveats, stated upfront. The size simulation below is a deliberately narrow diagnostic. Four limitations bound what it can establish, and we flag them before deriving anything:

1. **Size, not power or consistency.** The exercise asks only how often a procedure rejects a null that is true by construction. It speaks to the validity of inference (the standard errors), not to the consistency or efficiency of the point estimator.
2. **The placebo is spatially blind.** The data-generating process (DGP) permutes the *observed* climate series across spatial units. This preserves each series’ decadal persistence and marginal distribution but *destroys* the cross-unit spatial correlation of climate. The simulation therefore stresses only the serial/temporal dimension of dependence and is uninformative about whether a procedure is adequate for genuine spatial correlation (see §B.4).
3. **Asymptotic Conley inference.** The Conley standard errors are computed on the partialled-out (Frisch–Waugh) system, so the degrees of freedom are not adjusted for the absorbed fixed effects and trends. Inference is large-sample (normal); the resulting standard errors are, if anything, mildly anti-conservative.
4. **Short decadal panel.** The estimation panel is decadal ($T = 4$: 1981, 1991, 2001, 2011). “Trends” and “lags” are therefore measured in decades, and degrees of freedom in the time dimension are scarce.

B.1. The panel estimator

The estimating equation is the two-way fixed-effects panel of the main text, reproduced here for a spatial unit i in state r and decade t :

$$(A34) \quad Y_{irt} = \boldsymbol{\beta}'\mathbf{T}_{irt} + \boldsymbol{\phi}'\mathbf{P}_{irt} + \mu_i + \lambda_t + \gamma_r \cdot t + \varepsilon_{irt},$$

where $\mathbf{T}_{irt} = (T_{\min}, T_{\max})'_{irt}$ is the diurnal temperature pair (in surface-air units in the main text, wet-bulb here), $\mathbf{P}_{irt}$ is the contemporaneous rainfall control, μ_i are unit fixed effects, λ_t decade fixed effects, and $\gamma_r \cdot t$ a state-specific linear trend. The coefficient of interest is the temperature vector $\boldsymbol{\beta}$.

Collect the regressors of interest into $\mathbf{x}_{irt} = (\mathbf{T}'_{irt}, \mathbf{P}'_{irt})'$ and every other term—unit and decade dummies, the state trends, and any additional controls—into a nuisance design $\mathbf{z}_{irt}$, so that (A34) is $Y = \mathbf{x}'\boldsymbol{\theta} + \mathbf{z}'\boldsymbol{\eta} + \varepsilon$ with $\boldsymbol{\theta} = (\boldsymbol{\beta}', \boldsymbol{\phi}')'$. Let $\mathbf{M}_Z = \mathbf{I} - \mathbf{Z}(\mathbf{Z}'\mathbf{Z})^+\mathbf{Z}'$ be the annihilator of the stacked nuisance matrix $\mathbf{Z}$. By the Frisch–Waugh–Lovell (FWL) theorem the coefficient of interest solves the partialled-out least-squares problem

$$(A35) \quad \hat{\boldsymbol{\theta}} = (\tilde{\mathbf{X}}'\tilde{\mathbf{X}})^{-1}\tilde{\mathbf{X}}'\tilde{\mathbf{Y}}, \quad \tilde{\mathbf{X}} \equiv \mathbf{M}_Z\mathbf{X}, \quad \tilde{\mathbf{Y}} \equiv \mathbf{M}_Z\mathbf{Y},$$

i.e. one regresses the residualized outcome $\tilde{\mathbf{Y}}$ on the residualized regressors $\tilde{\mathbf{X}}$, where “residualized” means net of the unit/decade effects, the state trends, and the controls. Equation (A35) is an *algebraic identity*: it returns exactly the $\hat{\boldsymbol{\theta}}$ from estimating (A34) directly, because $\mathbf{M}_Z$ is the idempotent projection off $\text{col}(\mathbf{Z})$ and the partitioned normal equations separate. The point estimate

is thus invariant to how the variance is computed; only the standard error changes. This invariance is what lets us hold the estimate fixed and sweep the inference procedure below.

B.2. Variance estimators

Write the residualized residuals $\hat{\varepsilon} = \tilde{\mathbf{Y}} - \tilde{\mathbf{X}}\hat{\boldsymbol{\theta}}$ and $\mathbf{Q} = \tilde{\mathbf{X}}'\tilde{\mathbf{X}}$. All estimators are of the sandwich form $\mathbf{Q}^{-1}\hat{\boldsymbol{\Omega}}\mathbf{Q}^{-1}$ and differ only in the “meat” $\hat{\boldsymbol{\Omega}}$.

District clustering groups observations by district g and allows arbitrary within-district correlation across decades while assuming independence across districts:

$$(A36) \quad \hat{\boldsymbol{\Omega}}_{\text{cl}} = \sum_g \left(\sum_{irt \in g} \tilde{\mathbf{x}}_{irt} \hat{\varepsilon}_{irt} \right) \left(\sum_{irt \in g} \tilde{\mathbf{x}}_{irt} \hat{\varepsilon}_{irt} \right)'.$$

Conley spatial–temporal HAC replaces the cluster indicator with a kernel in geographic distance and time. For observations $a = (i, t)$ and $b = (j, s)$ with great-circle distance d_{ij} and decadal separation $|t - s|$,

$$(A37) \quad \hat{\boldsymbol{\Omega}}_{\text{Conley}} = \sum_a \sum_b K_{ab} \tilde{\mathbf{x}}_a \hat{\varepsilon}_a \hat{\varepsilon}_b \tilde{\mathbf{x}}_b', \quad K_{ab} = \underbrace{\mathbf{1}\{d_{ij} \leq \bar{d}\}}_{\text{spatial}} \cdot \underbrace{\left(1 - \frac{|t-s|}{L+1}\right)_+}_{\text{Bartlett, lag } L},$$

with spatial cutoff $\bar{d}$ (km) and time-lag cutoff L (decades). Two limiting cases clarify what (A37) does. With $\bar{d} \rightarrow 0$ the spatial term collapses and (A37) reduces to a within-unit Bartlett HAC; with $L \geq T - 1$ and $\bar{d}$ small the time kernel covers all within-unit pairs, so the Conley estimator approaches (A36) but with *triangular* rather than uniform weights on the within-unit lags—which is exactly why, in Table A3, the Conley standard error at 50 km and four lags is marginally smaller than, but otherwise tracks, the district-clustered one. In practice (A37) is evaluated on the FWL-residualized system $(\tilde{\mathbf{Y}}, \tilde{\mathbf{X}})$ of (A35), so the point estimate is unchanged and only the meat differs; inference is normal-based.

B.3. Specification selection by simulated size

Let $\mathcal{C} = \{c\}$ index the candidate procedures—combinations of a fixed-effects/trend structure (absorbed into $\mathbf{Z}$) and a variance estimator ((A36) or (A37) at various $\bar{d}, L$). We select the c whose empirical rejection rate of a true null is closest to the nominal 5%.

The DGP imposes the null by *randomization inference* on the regressor. For each replication $s = 1, \dots, S$ ($S = 1,000$) we draw a uniform random permutation π_s of the spatial units and construct a placebo climate field by reassigning each unit’s *entire* time series to another unit,

$$(A38) \quad \tilde{\mathbf{X}}_{irt}^{(s)} = \mathbf{X}_{\pi_s(i), t},$$

moving the $(T_{\min}, T_{\max}, \text{rain})$ block together so their joint structure is preserved. Because π_s is independent of the outcome, the placebo is unrelated to Y and the true temperature effect is $\boldsymbol{\beta} = \mathbf{0}$ by construction. For each candidate c we estimate (A34) with the placebo in place of the climate regressors and test the joint null on the temperature subvector, $H_0 : \boldsymbol{\beta} = \mathbf{0}$, via the Wald statistic

$$(A39) \quad W_c^{(s)} = \hat{\boldsymbol{\beta}}^{(s)'} [\mathbf{R} \hat{\mathbf{V}}_c^{(s)} \mathbf{R}']^{-1} \hat{\boldsymbol{\beta}}^{(s)},$$

where $\mathbf{R}$ selects the temperature coefficients and $\hat{\mathbf{V}}_c^{(s)}$ is procedure c ’s sandwich variance. The

empirical size is the rejection frequency

$$(A40) \quad \hat{\alpha}_c = \frac{1}{S} \sum_{s=1}^S \mathbf{1}\{p(W_c^{(s)}) < 0.05\},$$

and the preferred procedure satisfies $\hat{\alpha}_c \approx 0.05$. Table A3 reports $\hat{\alpha}_c$ across the Census-town outcomes and three climate regressor sets (the Census-district and ICRISAT panels behave identically). Unadjusted and heteroskedasticity-robust errors reject at three to four times the nominal rate because the placebo is highly persistent; district clustering restores $\hat{\alpha}_c \approx 0.05$, and adding state linear trends (or state $\times$ decade fixed effects) keeps it there, whereas district-specific trends over-absorb on the short panel.

TABLE A3. Placebo Monte-Carlo Size: Rejection Rates by Variance Estimator (Census Towns)

	Seasonal	Cultivators	Ag. labour	Non-ag.	Total workers
No adjustment (homoskedastic)	0.088	0.144	0.126	0.121	0.127
Heteroskedasticity-robust	0.073	0.106	0.098	0.093	0.080
Cluster by district	0.040	0.052	0.052	0.054	0.048
Cluster by district + district linear trends	0.045	0.041	0.044	0.058	0.036
Cluster by district + state linear trends	0.037	0.055	0.041	0.059	0.048
Cluster by district + state-by-year FE	0.039	0.048	0.047	0.064	0.048
Cluster by district + district-by-year FE	0.053	0.046	0.059	0.060	0.049
Cluster by state	0.083	0.093	0.068	0.077	0.073
Conley spatial HAC (50 km)	0.154	0.192	0.178	0.168	0.154
Conley spatial HAC (100 km)	0.150	0.209	0.182	0.180	0.158
Conley spatial HAC (200 km)	—	—	—	—	—
Conley spatial HAC (50 km, 2 lags)	0.074	0.106	0.106	0.099	0.110
Conley spatial HAC (50 km, 4 lags)	0.043	0.049	0.050	0.051	0.049
Replications	1000				

Notes: Empirical rejection rate $\hat{\alpha}_c$ of a true null at the nominal 5% level, from 1,000 placebo replications that permute the climate series across spatial units (A38), for each variance estimator (rows) and Census-town outcome (columns). The bold row is the preferred specification—district clustering with state-specific linear trends—which restores nominal size; the Census-district and ICRISAT panels behave identically. “—” marks cells the Conley estimator at that radius does not return.

B.4. What the size simulation does and does not identify

The placebo (A38) relocates whole series across units. It therefore *preserves* (i) each series’ marginal distribution, (ii) its within-unit serial persistence—a donor district’s own trending decadal trajectory—and (iii) the common secular time profile shared across units; but it *eliminates* the spatial correlation of climate, since neighboring units receive unrelated donors. The only genuine dependence left in the placebo is within-unit and serial. Two consequences follow. First, the spatial radius $\bar{d}$ in (A37) has almost nothing to correct, which is why the rejection rates in Table A3 are nearly flat across 50–200 km; the temporal lag L does the work. Second, district clustering and a Conley HAC with enough decadal lags coincide in size *because both absorb exactly the within-unit serial dependence that this DGP contains*—their agreement is mechanical and confined to the temporal dimension. The simulation cannot rank the two on spatial adequacy, the dimension on which they genuinely differ in the real data (where heat shocks are spatially correlated across nearby districts and district clustering, unlike Conley, assumes cross-unit independence). A sharper experiment, left to future work, would permute climate in spatial blocks (e.g. at the region or state level) so that the placebo *retains* spatial correlation; under that DGP district clustering would be expected

to over-reject while a Conley correction held size. We therefore read the present diagnostic as establishing the need to handle serial persistence—which the preferred estimator does—and report Conley standard errors as a complementary check on the spatial dimension rather than as a procedure the size test has validated against it.

B.5. Why state-specific linear trends

The term $\gamma_r \cdot t$ in (A34) is an economic restriction, not only a variance device. Over 1981–2011 Indian states followed divergent and largely smooth trajectories of structural transformation, driven by state-level policy regimes, infrastructure, and agro-ecological endowments, and these trajectories are correlated with the slow secular warming that the decadal moving-average climate also tracks. Absorbing a state-specific linear trend purges this confounding low-frequency co-movement, so that β is identified from deviations of a unit’s outcome from its state’s own linear path, comparing units within their local economic geography under a common decadal climate trajectory. It is the least aggressive control that accomplishes this: the more saturated alternative, state $\times$ decade fixed effects, additionally discards *all* within-state-decade variation—including the spatially correlated climate signal of interest—and so risks absorbing the treatment itself; the FE-structure robustness in Tables A34–A37 shows the estimates are nonetheless stable across this choice. The size simulation (§B.3) and this argument are complementary: the state trend is the specification that both *restores nominal size* against the persistent placebo and *carries a substantive interpretation*, which a purely statistical degrees-of-freedom adjustment would not.

Appendix C. Data Sources and Construction

I harmonize four sources onto a fixed spatial frame: the India Meteorological Department (IMD) gridded climate, the ICRISAT District Level Database, the Indian Census (district and town tables), and the NSS Employment and Unemployment schedules. The unit of analysis is the 2011 Census district (and, for the town exhibits, the Census town). This appendix documents the concordance, the spatial assignment of climate, and the construction of each sample.

C.1. Reproduction and data lineage

The project is reproduced by a four-stage pipeline, raw $\rightarrow$ build $\rightarrow$ analysis $\rightarrow$ exhibits, each stage writing to a parallel data directory:

1. **Build** (`code/01_build/` $\rightarrow$ `data/build/output/`): cleans each raw source into a harmonized panel on the 2011 frame—one subdirectory per source (`imd`, `dchb_district`, `dchb_town`, `icrisat`, `ipums`, plus the GIS layers `HydroSHEDS`, `ind_adm_bounds`). Climate exposure is assigned here by inverse-distance-squared weighting of surrounding IMD grid points (§C.3).
2. **Prep** (`code/final/01_prep/` $\rightarrow$ `data/final/01_prep/`): merges climate onto each outcome panel to produce the four analysis datasets used throughout—`districtPCA_climate` (Census districts), `townPCA_climate` (Census towns), `icrisat_climate` (ICRISAT districts), and the IPUMS–NSS panel.
3. **Estimation** (`code/final/03_reg/` $\rightarrow$ `data/final/03_reg/`): the `dchbreg` engine runs the panel (eq. A34) and long-difference estimators, sweeping outcome $\times$ climate specification $\times$ fixed-effect structure $\times$ standard-error family, and stores one tidy row per cell. Summary statistics and the Monte-Carlo size simulations of §B.3 live alongside in `code/final/02_sum/`.
4. **Exhibits** (`code/final/04_tab/`, `05_fig/`, `06_oa/` $\rightarrow$ `tex/arxiv/`): render the tables and figures of the paper and this appendix from the tidy estimate frames.

The analysis datasets trace to raw sources as follows. The Census-district panel and the ICRISAT panel each combine their respective build output (`data/build/output/{dchb_district,icrisat}`) with the IMD climate grid; the town panel additionally merges river proximity (`HydroSHEDS`) and canal irrigation; the IPUMS panel merges the NSS employment schedules onto district climate. Each prep do-file (`code/final/01_prep/{dchb,icrisat,ipums}/`) is named for the analysis dataset it produces, so the lineage from any regression back to raw is recoverable by following the build-output inputs it loads.

Table A4 draws this lineage at the file level: for each of the four analysis datasets, and for the shared climate spine, it traces the raw inputs through the build and prep scripts to the clean indicators the estimates use.

C.2. District concordance

India’s district boundaries change across rounds through splits and, less often, mergers, so I map every Census round (1961–2011) onto a fixed 2011 frame using a name-and-code crosswalk (the `indiabridge` utility, keyed on the 1981, 2001, and 2011 district codes). Tables for 1961–1991 come from the Vanneman compilation and for 2001–2011 from the digitized Primary Census Abstract. The concordance is a least-common-unit (inner) join: a district enters the panel only if it matches cleanly across all sources. I do not apportion population between split parents and children; instead, historical districts that cannot be uniquely assigned to a single 2011 unit (for example composite Vanneman records spanning more than one successor) are dropped rather

TABLE A4. Input data map: raw sources to analysis indicators, by sample

Analysis dataset (unit)	Raw inputs	Build → prep scripts	Clean indicators
<i>Climate spine—shared, and joined to every sample below by inverse-distance-squared weighting of surrounding IMD grid points (§C.3).</i>			
imd grid (climate)	73 daily gridded CSVs from IMD: air temperature (1°) and rainfall (0.25°), 1951–2023	01_build/imd/03-09 → 01_prep/imd/01-02	T_{min} , T_{max} , T_{bar} , degree-days, wet-bulb, rain; growing-season (Jun–Feb) means and 10-yr moving averages
Districts districtPCA_ climate (Census district)	Vanneman PCA .dta (1961–91); PCA-2001 and PCA-2011 .dta; india_bridge.dta concordance; Survey-of-India and historical district shapefiles; 1961 district-area CSV	01_build/ dchb_district/ 00-05 → 01_prep/dchb/ districtPCA_climate	Worker shares {ag. labor, cultivator, seasonal, non-agriculture} × {total, rural, urban}; log workers; log population
Towns townPCA_ climate (+_LD) (Census town)	Town-2011 shapefile; townbridge_all cross-decade crosswalk; per-decade digitized DCHB Primary Census Abstract (CSV/XLSX, 1961–2011); statement A-4 area and civic status	01_build/ dchb_town/ 00-04 → 01_prep/dchb/ townPCA_climate (_LD)	Four Census worker shares; total workers; town area; civic (statutory/Census-town) status
ICRISAT icrisat_ climate (district)	district_crops_APY.csv; district_Apportioned Identifiers.csv; icrisat_cen2011_match.csv	01_build/ icrisat/00-01 → 01_prep/icrisat/ icrisat_climate	Cultivated area, production, yield, harvest price × six grains (rice, wheat, maize, sorghum, barley, millets)
IPUMS/NSS ipums_ climate (district)	ipumsi_00001.dta: NSS Employment–Unemployment rounds 1987/1999/2009 via IPUMS; geo2 shapefile	01_build/ ipums/00-02 → 01_prep/ipums/ ipums_climate	Cultivator, ag. labor, non-agriculture (informal, formal) shares; urban subsample

Notes: Each prep do-file is named for the analysis dataset it writes; the build subdirectory cleans each raw source onto the 2011 Census frame before climate is assigned. Climate is identical in construction across samples (the shared spine, top row) and is merged onto each unit by its centroid. Paths are relative to code/; ranges such as 00-05 denote the numbered do-files run in sequence.

than divided, trading coverage for boundary stability. Eight states are excluded for want of a usable 1981 identifier (Arunachal Pradesh, Goa–Daman–Diu, Manipur, Meghalaya, Nagaland, Tripura, Sikkim, and Mizoram). A handful of district-decades are missing by design and coded as such rather than zero: Assam in 1981 and Jammu & Kashmir in 1991 (Census not conducted), Calcutta in 1961–1991, and the Andaman & Nicobar Islands in 1961. The harmonized panel contains 300 districts; the decadal panel sample is 1,187 district-decades and the long-difference sample is 293 districts. Worker categories are reconciled across the 1981/1991 and 2001/2011 definitions: total workers are main plus marginal workers, agricultural workers are cultivators plus agricultural labourers, and non-agricultural workers are the residual of main workers. All shares are winsorized at the 1st and 99th percentiles within decade.

A known enumeration change affects the marginal-worker category—the “seasonal” margin in the main text. The 2001 Census captured marginal and casual work far more completely than 1991, so the marginal-worker share roughly doubles across the 1991–2001 boundary (rural district seasonal shares rise from about 0.12 to 0.26, and balanced-town shares from about 0.03 to 0.12; Table A6). Two features contain this break. First, the *headline* divergence is identified off the *main*-worker categories—agricultural wage labor and self-cultivation—which are defined consistently throughout and are not reclassified by the marginal-worker change; only the seasonal-sink channel draws on the marginal count, and there the night/day split is the weaker of the reported results. Second, the common component of the 2001 shift is a level break shared across districts within a round, which the decade fixed effects in the panel absorb; the long difference, which straddles the 1991–2001 boundary, is the more exposed design, and its estimates are reported alongside the panel throughout so the two can be read against one another. Worker shares are therefore best read as main-worker reallocations, with the seasonal margin as the flexible—and more definition-sensitive—residual.

C.3. Climate exposure and IMD assignment

Daily temperature and precipitation come from the IMD gridded products (air temperature at 1°, precipitation at 0.25°, 1951–2023). I assign grid points to each district or town centroid by inverse-distance-squared weighting: for every centroid I take all grid points within 200 km (100 km for towns), weight each by $1/d^2$ normalized to sum to one, and form the weighted average. The cutoffs match the distance at which the residual spatial correlation of the outcomes decays (about 150–200 km for districts and 100 km for towns). Because the weights decay with the square of distance, points beyond roughly 100 km contribute negligibly, so the assignment is effectively local and insensitive to the outer cutoff—consistent with the residual correlograms reported in the main text, whose spatial correlation decays within about 100 km. The harmonized grid comprises 362 points. Following Liu, Shamdasani, and Taraz (2023), decadal exposure is the mean over the growing seasons of the preceding ten years, so that climate and outcomes share a common ten-year horizon at each Census. The growing season spans June–February (the kharif and rabi cropping cycles); months are tagged kharif (June–October), rabi (November–February), and lean (March–May), and the lean months are excluded from the growing-season aggregates. The baseline margins are the growing-season daily minimum and maximum surface-air temperatures, T_{min} and T_{max} (their wet-bulb counterparts T_{wmin} and T_{wmax} are the default in this appendix), entered simultaneously. As a scalar alternative I construct growing-season degree days, accumulating daily mean temperature between 8 and 32 °C (Blakeslee and Fishman 2015; Schlenker, Hanemann, and C. 2006), which average over the diurnal cycle and so collapse the night and day margins into one. Standard errors are clustered at the district level throughout; the main text also reports spatially-corrected (Conley 1999) standard errors as a check on residual spatial autocorrelation.

C.4. Wet-bulb temperatures

The humidity-adjusted measures use *no external humidity series*: relative humidity is recovered from the daily temperature pair itself and converted to a wet-bulb temperature by the empirical formula of Stull (2011). Daily actual and saturation vapour pressures are computed from the Tetens equation, $e(T) = 6.1078 \exp[17.269 T / (237.3 + T)]$, using the daily minimum as a proxy for the dewpoint: $e_a = e(T_{min})$ and $e_s = e(T_{max})$, so daytime relative humidity is $RH = 100 e_a / e_s$, capped at 100%. The daytime wet-bulb temperature then follows Stull (2011),

$$T_{wmax} = T_{max} \arctan[0.151977 (RH + 8.313659)^{1/2}] + \arctan(T_{max} + RH) - \arctan(RH - 1.676331) \\ + 0.00391838 RH^{3/2} \arctan(0.023101 RH) - 4.686035,$$

while $T_{wmin} = T_{min}$ (at the daily minimum the dewpoint proxy implies $RH \approx 100\%$, where wet-bulb and dry-bulb temperatures coincide). The wet-bulb minimum and maximum are aggregated to growing-season decadal means exactly as the surface-air measures. Because T_{wmin} and T_{wmax} are deterministic functions of T_{min} and T_{max} , the wet-bulb specifications are a nonlinear robustness re-expression of the surface-air pair rather than an independent humidity measurement, and are reported as a supplementary check.

C.5. Agricultural data: ICRISAT

Cultivated area, production, yields, and harvest prices come from the ICRISAT District Level Database (annual, 1966–present), averaged over the same ten-year windows preceding each Census. ICRISAT’s agro-economic districts are not nested in Census boundaries, so I match them to 2011 Census districts in stages, by decreasing confidence: a set of manual overrides for known splits and renamings, then exact normalized-name matches, then substring matches, then fuzzy (edit-distance) matches at a 0.6 similarity threshold; Telangana districts are folded into Andhra Pradesh to match the pre-2014 frame. Of 313 ICRISAT districts, 237 match exactly, 18 by substring, 33 by fuzzy rule, and 25 by manual override, with 0 unmatched. The estimation panel covers 311 districts. Crop outcomes are built for rice, wheat, maize, sorghum, barley, and millets, and for a pooled cereal index.

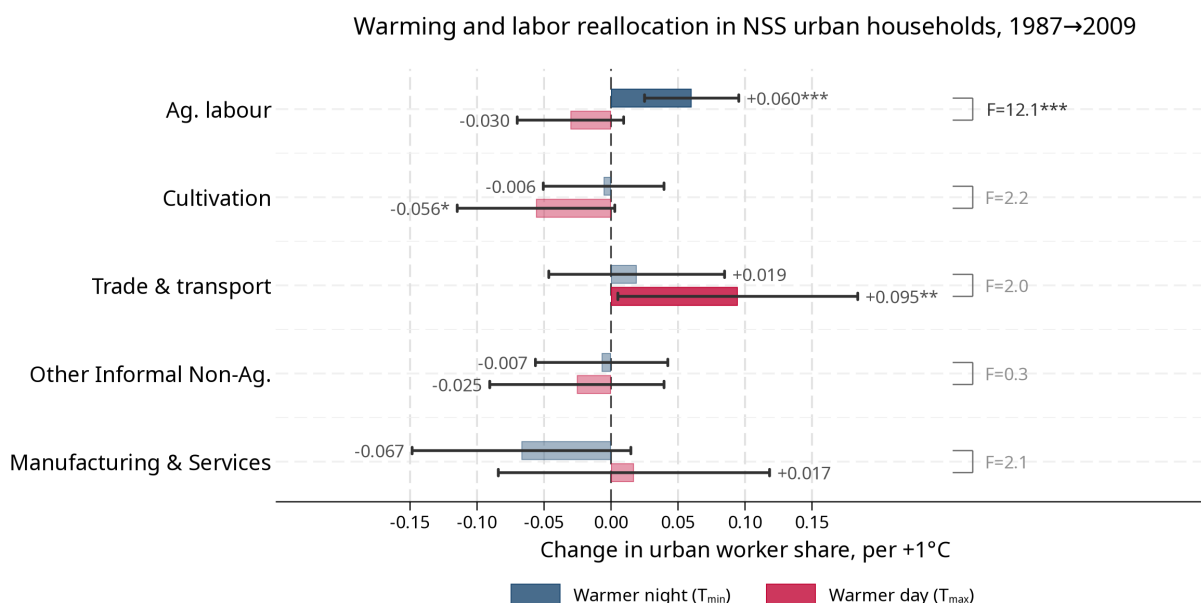
C.6. Town panel

The town data are digitized from the Primary Census Abstracts, 1961–2011. A Census town must satisfy the standard criteria (population at least 4,000, at least 75% of male workers outside agriculture, and density at least 400 persons per km²). Towns are not a stable universe: places enter as large villages “graduate” to town status and, occasionally, exit. I therefore use two samples. The town *panel* uses the full set of towns observed from 1981 onward (8,786 towns, 16,215 town-decades); the town *long-difference* restricts to the balanced set present in every round since 1981 (2,969 towns), so the endpoint comparison reflects intensive-margin adjustment within continuing towns rather than the reclassification of new ones. Because the appearance of new towns is itself partly an outcome of growth, the balanced long-difference is the more conservative design for this composition concern, and the panel and balanced results are reported alongside one another. Occupations use the same four-way split as the district data (cultivators, agricultural labourers, non-agricultural workers, and marginal workers).

C.7. NSS employment surveys (IPUMS)

To resolve the destination sector of displaced labour—which the Census’s 1981 occupation definitions do not—I use IPUMS-International extracts of the NSS Employment and Unemployment schedules (Ruggles et al. 2024), for rounds spanning 1987, 1999, and 2009. The unit is the harmonized sub-national region (geo2), restricted to urban households as a benchmark to the Census town sample (416 regions, 1,237 region-rounds). These rounds are *not* merged onto Census years: each round is treated as its own observation and paired with climate averaged over its own preceding ten-year window (for example 1977–1986 for the 1987 round), so the worker-share levels are never assumed to proxy a specific Census year. Occupations are grouped into five mutually exclusive sectors that partition the urban workforce: agriculture (wage labour and self-cultivation), trade and transport, other informal work (construction, mining, utilities, hospitality, and personal services), and formal manufacturing and services. NSS results appear only as suggestive evidence. Figure A1 visualises the sectoral destination; Table A5 reports the full coefficients.

FIGURE A1. Where heat-displaced urban labour goes (NSS/IPUMS)



Notes: Suggestive evidence for future work. Panel response of each urban worker share to a 1°C rise in the growing-season nighttime minimum (T_{min} , navy) and daytime maximum (T_{max} , cranberry), from three rounds of the NSS Employment–Unemployment schedules (IPUMS-International, 1987–2009), restricted to urban households. The five shares partition the urban workforce. Spikes are 95% confidence intervals; full-colour bars are significant at 5%. The brace on each sector reports the night-versus-day equality test $F(T_{min} = T_{max})$, significant only for agricultural wage labour. Full coefficients are in Table A5.

TABLE A5. NSS Urban Households: Sectoral Destination of Displaced Labour

	Agriculture		Informal Non-Agriculture		Formal
	Ag. Labour (1)	Culti- vation (2)	Trade, Transport & Comm. (3)	Other Informal (4)	Manuf. & Services (5)
T_{min}	0.060*** (0.0179) [0.0169]	-0.006 (0.0230) [0.0222]	0.019 (0.0335) [0.0314]	-0.007 (0.0252) [0.0242]	-0.067 (0.0416) [0.0409]
T_{max}	-0.030 (0.0203) [0.0194]	-0.056* (0.0300) [0.0288]	0.095** (0.0457) [0.0436]	-0.025 (0.0332) [0.0321]	0.017 (0.0516) [0.0507]
Observations	1,163	1,163	1,163	1,163	1,163
Dep Var. Mean	0.04	0.07	0.31	0.14	0.44
F-stat ($T_{min} = T_{max}$)	12.08	2.17	2.03	0.26	2.07
Rainfall Control	Y	Y	Y	Y	Y
State Linear Trend	Y	Y	Y	Y	Y

Notes: Urban worker shares in the NSS Employment–Unemployment rounds (IPUMS-International, 1987–2009), restricted to urban households and classified into five mutually exclusive sectors that partition the urban workforce: agricultural wage labour and self-cultivation (cultivators and unpaid family workers); two informal non-agricultural sectors—trade/transport/communication, and other informal work (construction, mining, utilities, hospitality, and personal and household services)—grouped as urban casual labour; and formal manufacturing and high-skill services. The NSS urban frame differs from the Census town definition—it retains the peri-urban self-cultivation a Census town (by construction over 75% non-agricultural) excludes—so the two urban measures agree on the agricultural wage-labour margin but not on self-cultivation. Decadal panel with district and decade fixed effects and a state $\times$ decade linear trend, and precipitation controls. Parentheses report district-clustered standard errors; brackets report Conley spatial–temporal HAC standard errors (50 km, four decadal lags). The F -statistic tests $T_{min} = T_{max}$. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

C.8. Rainfall control

Every specification controls for precipitation. Rainfall is taken from the same IMD gridded daily product (0.25°) and assigned to districts and towns with the identical inverse-distance-squared scheme, then aggregated to growing-season totals and averaged over the same preceding-decade window as temperature. No alternative gridded product (APHRODITE, CRU) is used. Because rainfall covaries with both temperature margins, controlling for it over the matched window is what isolates the partial temperature effects.

C.9. Summary statistics

Table A6 reports means and standard deviations of the main outcomes and of the temperature and rainfall regressors, separately for the decadal panel and the long-difference samples, with district and observation counts for each.

TABLE A6. Summary Statistics: Panel and Long-Difference Samples

	Decadal panel (levels, 1981–2011)	Long difference (Δ 2001–11 vs. 1981–91)
Cultivators (share)	0.323 (0.158)	-0.139 (0.080)
Agricultural labourers (share)	0.180 (0.111)	-0.049 (0.057)
Seasonal / marginal (share)	0.173 (0.111)	0.139 (0.086)
Non-agricultural (share)	0.324 (0.176)	0.048 (0.070)
Nighttime T_{min} (°C)	17.95 (2.39)	0.28 (0.24)
Daytime T_{max} (°C)	29.71 (1.95)	0.15 (0.19)
Rainfall (growing-season mean)	1.0 (0.5)	0.0 (0.1)
Districts	300	299
Observations	1,187	299

Notes: Means with standard deviations in parentheses, for the decadal district panel (1981–2011) and the long-difference cross-section (1981–91 vs. 2001–11). Outcomes are shares of total district workers; temperature and rainfall are growing-season ten-year moving averages. District and observation counts are reported at the foot of the table; these are descriptive samples, and the estimation samples are marginally smaller after dropping observations with missing climate covariates.

Appendix D. Robustness to Climate Measurement

TABLE A7. ICRISAT Districts, Long Difference: Alternative Climate Measures

	Cultivated Area (1)	Yield Index (2)	Field Wage (3)	Price Index (4)
<i>Wet-bulb T_{wmin}/T_{wmax}, growing season (baseline)</i>				
ΔT_{wmin}	0.289 (0.2017) [0.1954]	0.357** (0.1668) [0.1601]	0.144 (0.1904) [0.1849]	-0.118 (0.1013) [0.0981]
ΔT_{wmax}	-1.715*** (0.6231) [0.6034]	-0.626 (0.5064) [0.4897]	-0.675** (0.3403) [0.3204]	0.138 (0.2002) [0.1943]
F-stat ($\Delta T_{wmin} = \Delta T_{wmax}$)	6.15	2.23	2.57	0.76
Observations	311	303	259	294
<i>Dry-bulb T_{min}/T_{max}, growing season</i>				
ΔT_{min}	-0.584*** (0.1654) [0.1599]	0.052 (0.1362) [0.1320]	-0.193* (0.1041) [0.1048]	-0.045 (0.0493) [0.0515]
ΔT_{max}	-0.607*** (0.2204) [0.2137]	-0.178 (0.1819) [0.1758]	-0.212* (0.1227) [0.1151]	0.061 (0.0750) [0.0730]
F-stat ($\Delta T_{min} = \Delta T_{max}$)	0.03	4.56	0.02	1.81
Observations	311	303	259	294
<i>Degree days, growing season</i>				
Δ Degree Days	-1.133*** (0.3658) [0.3553]	-0.072 (0.3021) [0.2928]	-0.410** (0.1718) [0.1705]	-0.033 (0.0951) [0.0979]
Observations	311	303	259	294
<i>Degree days, calendar year</i>				
Δ Degree Days	-1.097*** (0.4001) [0.3880]	-0.080 (0.3278) [0.3174]	-0.362** (0.1764) [0.1764]	-0.021 (0.0994) [0.1034]
Observations	311	303	259	294
Dep. var. mean	-0.076	0.368	1.919	1.364

Notes: Each block re-estimates the ICRISAT long-difference specification (Table 1) replacing the climate regressors. Outcomes are log levels for the cereal index (rice, wheat, barley, maize, millets, sorghum). Long-differenced cross-section: outcomes and climate averaged over 1981–91 (baseline) and 2001–11 (endline), then differenced. All specifications include state fixed effects. Temperatures are wet-bulb unless noted, constructed as a moving average of the preceding decade's growing seasons (June–February), as inverse-distance-squared weighted averages of surrounding IMD grid points (see Appendix C.3). All specifications control for total precipitation over the same window unless noted. Standard errors in parentheses, clustered at the district level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

TABLE A8. Census Districts, Long Difference: Alternative Climate Measures

	Seasonal (1)	Ag. Labour (2)	Cultivation (3)	Non-Agriculture (4)
<i>Wet-bulb T_{wmin}/T_{wmax}, growing season (baseline)</i>				
ΔT_{wmin}	-0.147*** (0.0440) [0.0430]	0.159*** (0.0329) [0.0329]	-0.009 (0.0512) [0.0497]	0.001 (0.0469) [0.0451]
ΔT_{wmax}	0.144* (0.0863) [0.0848]	-0.165** (0.0652) [0.0649]	-0.031 (0.1004) [0.0974]	0.049 (0.0968) [0.0936]
F-stat ($\Delta T_{wmin} = \Delta T_{wmax}$)	5.42	11.95	0.02	0.12
Observations	293	293	293	293
<i>Dry-bulb T_{min}/T_{max}, growing season</i>				
ΔT_{min}	-0.074*** (0.0269) [0.0254]	0.075*** (0.0209) [0.0202]	-0.024 (0.0308) [0.0295]	0.026 (0.0281) [0.0271]
ΔT_{max}	0.049 (0.0334) [0.0327]	-0.062** (0.0246) [0.0247]	-0.008 (0.0376) [0.0365]	0.020 (0.0365) [0.0353]
F-stat ($\Delta T_{min} = \Delta T_{max}$)	12.34	26.07	0.16	0.03
Observations	293	293	293	293
<i>Degree days, growing season</i>				
Δ Degree Days	-0.086* (0.0484) [0.0463]	0.080** (0.0402) [0.0390]	-0.043 (0.0565) [0.0542]	0.053 (0.0529) [0.0512]
Observations	293	293	293	293
<i>Degree days, calendar year</i>				
Δ Degree Days	-0.123** (0.0503) [0.0478]	0.111*** (0.0410) [0.0396]	-0.047 (0.0583) [0.0560]	0.063 (0.0559) [0.0539]
Observations	293	293	293	293
Dep. var. mean	0.141	-0.049	-0.141	0.049

Notes: Each block re-estimates the district worker-share long-difference specification (Table 2) replacing the climate regressors. Outcomes are shares of total district workers. Long-differenced cross-section: outcomes and climate averaged over 1981–91 (baseline) and 2001–11 (endline), then differenced. All specifications include state fixed effects. Temperatures are wet-bulb unless noted, constructed as a moving average of the preceding decade’s growing seasons (June–February), as inverse-distance-squared weighted averages of surrounding IMD grid points (see Appendix C.3). All specifications control for total precipitation over the same window unless noted. Standard errors in parentheses, clustered at the district level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

TABLE A9. Census Towns, Long Difference: Alternative Climate Measures

	Seasonal (1)	Ag. Labour (2)	Cultivation (3)	Non-Agriculture (4)
<i>Wet-bulb T_{wmin}/T_{wmax}, growing season (baseline)</i>				
ΔT_{wmin}	-0.042** (0.0209) [0.0231]	0.003 (0.0225) [0.0242]	-0.048** (0.0220) [0.0237]	0.087*** (0.0308) [0.0331]
ΔT_{wmax}	0.051 (0.0390) [0.0432]	0.119*** (0.0419) [0.0463]	0.079** (0.0398) [0.0414]	-0.249*** (0.0612) [0.0648]
F-stat ($\Delta T_{wmin} = \Delta T_{wmax}$)	2.63	3.47	4.50	14.29
Observations	2,969	2,969	2,969	2,969
<i>Dry-bulb T_{min}/T_{max}, growing season</i>				
ΔT_{min}	-0.016 (0.0119) [0.0123]	0.063*** (0.0116) [0.0123]	-0.009 (0.0108) [0.0111]	-0.038** (0.0176) [0.0187]
ΔT_{max}	0.017 (0.0144) [0.0162]	0.041*** (0.0157) [0.0172]	0.022 (0.0144) [0.0148]	-0.078*** (0.0225) [0.0239]
F-stat ($\Delta T_{min} = \Delta T_{max}$)	4.06	1.61	3.38	2.78
Observations	2,969	2,969	2,969	2,969
<i>Degree days, growing season</i>				
Δ Degree Days	-0.005 (0.0210) [0.0225]	0.116*** (0.0208) [0.0229]	-0.002 (0.0183) [0.0186]	-0.110*** (0.0317) [0.0339]
Observations	2,969	2,969	2,969	2,969
<i>Degree days, calendar year</i>				
Δ Degree Days	-0.020 (0.0216) [0.0231]	0.118*** (0.0215) [0.0238]	-0.014 (0.0195) [0.0200]	-0.084** (0.0333) [0.0368]
Observations	2,969	2,969	2,969	2,969
Dep. var. mean	0.102	-0.053	-0.058	0.010

Notes: Each block re-estimates the town worker-share long-difference specification (Table 3) replacing the climate regressors. Sample is the balanced panel of towns existing since 1981. Long-differenced cross-section: outcomes and climate averaged over 1981–91 (baseline) and 2001–11 (endline), then differenced. All specifications include state fixed effects. Temperatures are wet-bulb unless noted, constructed as a moving average of the preceding decade's growing seasons (June–February), as inverse-distance-squared weighted averages of surrounding IMD grid points (see Appendix C.3). All specifications control for total precipitation over the same window unless noted. Standard errors in parentheses, clustered at the district level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

TABLE A10. Census Towns, Panel: Alternative Climate Measures

	Seasonal (1)	Ag. Labour (2)	Cultivation (3)	Non-Agriculture (4)
<i>Wet-bulb T_{wmin}/T_{wmax}, growing season (baseline)</i>				
T_{wmin}	-0.023** (0.0113) [0.0106]	0.030** (0.0118) [0.0100]	-0.017* (0.0094) [0.0084]	0.008 (0.0160) [0.0148]
T_{wmax}	0.046** (0.0184) [0.0177]	-0.025 (0.0190) [0.0165]	0.016 (0.0133) [0.0125]	-0.037 (0.0258) [0.0237]
F-stat ($T_{wmin} = T_{wmax}$)	5.91	3.37	2.15	1.29
Observations	16,215	16,215	16,215	16,215
<i>Dry-bulb T_{min}/T_{max}, growing season</i>				
T_{min}	0.002 (0.0063) [0.0061]	0.017*** (0.0052) [0.0051]	-0.008** (0.0042) [0.0044]	-0.012 (0.0084) [0.0082]
T_{max}	0.016** (0.0069) [0.0065]	-0.007 (0.0069) [0.0060]	0.004 (0.0046) [0.0044]	-0.012 (0.0094) [0.0086]
F-stat ($T_{min} = T_{max}$)	2.58	7.22	2.83	0.00
Observations	16,215	16,215	16,215	16,215
<i>Degree days, growing season</i>				
Degree Days	0.016 (0.0100) [0.0097]	0.016* (0.0082) [0.0081]	-0.008 (0.0054) [0.0063]	-0.025* (0.0131) [0.0126]
Observations	16,215	16,215	16,215	16,215
<i>Degree days, calendar year</i>				
Degree Days	0.009 (0.0088) [0.0088]	0.012 (0.0084) [0.0079]	-0.005 (0.0048) [0.0056]	-0.016 (0.0132) [0.0122]
Observations	16,215	16,215	16,215	16,215
Dep. var. mean	0.095	0.094	0.077	0.732

Notes: Each block re-estimates the town worker-share decadal-panel specification replacing the climate regressors. Decadal panel (1981–2011) with unit and decade fixed effects and a state $\times$ decade linear trend. Temperatures are wet-bulb unless noted, constructed as a moving average of the preceding decade's growing seasons (June–February), as inverse-distance-squared weighted averages of surrounding IMD grid points (see Appendix C.3). All specifications control for total precipitation over the same window unless noted. Standard errors in parentheses, clustered at the district level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

TABLE A11. NSS Urban Households, Panel: Alternative Climate Measures

	Agriculture (1)	Constr. & Mining (2)	Manufacturing (3)	Services (4)
<i>Wet-bulb T_{wmin}/T_{wmax}, growing season (baseline)</i>				
T_{wmin}	0.160*** (0.0614) [0.0600]	0.060 (0.0408) [0.0395]	-0.064 (0.0485) [0.0467]	-0.155** (0.0725) [0.0698]
T_{wmax}	-0.196* (0.1092) [0.1052]	-0.105 (0.0799) [0.0770]	0.091 (0.0880) [0.0846]	0.210 (0.1321) [0.1281]
F-stat ($T_{wmin} = T_{wmax}$)	4.63	1.99	1.41	3.42
Observations	1,163	1,163	1,163	1,163
<i>Dry-bulb T_{min}/T_{max}, growing season</i>				
T_{min}	0.055* (0.0311) [0.0303]	0.006 (0.0217) [0.0211]	-0.014 (0.0282) [0.0274]	-0.047 (0.0388) [0.0374]
T_{max}	-0.086** (0.0399) [0.0385]	-0.035 (0.0300) [0.0291]	0.051 (0.0341) [0.0330]	0.071 (0.0489) [0.0475]
F-stat ($T_{min} = T_{max}$)	9.00	1.77	2.63	4.38
Observations	1,163	1,163	1,163	1,163
<i>Degree days, growing season</i>				
Degree Days	-0.019 (0.0545) [0.0522]	-0.013 (0.0399) [0.0385]	0.029 (0.0473) [0.0460]	0.004 (0.0660) [0.0640]
Observations	1,163	1,163	1,163	1,163
<i>Degree days, calendar year</i>				
Degree Days	-0.037 (0.0559) [0.0526]	-0.017 (0.0384) [0.0367]	0.034 (0.0478) [0.0467]	0.020 (0.0671) [0.0648]
Observations	1,163	1,163	1,163	1,163
Dep. var. mean	0.112	0.094	0.170	0.624

Notes: Each block re-estimates the NSS/IPUMS urban sector-share panel specification replacing the climate regressors. Outcomes are shares of urban household employment by sector (NSS rounds 1987–2009). Decadal panel (1981–2011) with unit and decade fixed effects and a state $\times$ decade linear trend. Temperatures are wet-bulb unless noted, constructed as a moving average of the preceding decade’s growing seasons (June–February), as inverse-distance-squared weighted averages of surrounding IMD grid points (see Appendix C.3). All specifications control for total precipitation over the same window unless noted. Standard errors in parentheses, clustered at the district level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Appendix E. Robustness to the Outcome Denominator: Per-Capita Worker Rates

The worker shares in the main text are denominated by the local workforce: each share is a category’s worker count divided by total workers, and the four mutually exclusive categories sum to one. Table A12 re-estimates the long difference with outcomes denominated instead by *population*—per-capita worker rates (a category’s count divided by population), which sum to the work-participation rate (column 5) rather than to one. The district agricultural wage-

labour divergence is unchanged: warmer nights raise, and warmer days lower, the per-capita ag-labour rate—sharply for the rural population ($F = 33$) and faintly for the urban ($F = 2$), the same amplitude gradient as the worker-share results in Figure 2. The per-capita lens also surfaces a participation margin the share cannot: warmer nights raise the rural work-participation rate (column 5, $+0.033$, $F = 7.4$ against a flat daytime coefficient), so the nighttime pull into agricultural wage-labour reflects both a recomposition of the existing workforce and net entry into it. In Census towns the agricultural wage-labour result likewise replicates; the one margin that attenuates under the population denominator is the non-agricultural share, consistent with part of the town worker-share contraction operating through rising agricultural participation rather than a fall in non-agricultural employment per capita.

TABLE A12. Per-Capita Worker Rates, Long Difference (1981–2011)

	Per-capita worker rate (share of population)				
	Ag. Labour (1)	Cultivator (2)	Seasonal (3)	Non- Agriculture (4)	Work Particip. (5)
<i>Panel A. Districts — Total population</i>					
ΔT_{min}	0.028 (3.20)	0.013 (0.95)	-0.039 (-3.12)	0.016 (1.60)	0.020 (1.50)
ΔT_{max}	-0.027 (-2.67)	-0.013 (-0.84)	0.016 (1.03)	0.005 (0.35)	-0.016 (-0.99)
F ($\Delta T_{min} = \Delta T_{max}$)	25.43	2.36	11.44	0.59	3.65
p-value	<0.001	0.126	<0.001	0.444	0.057
Observations			293		
<i>Panel B. Districts — Rural population</i>					
ΔT_{min}	0.039 (3.83)	0.020 (1.30)	-0.032 (-2.34)	0.007 (0.93)	0.033 (2.31)
ΔT_{max}	-0.032 (-2.84)	-0.009 (-0.53)	0.020 (1.20)	-0.004 (-0.30)	-0.020 (-1.19)
F ($\Delta T_{min} = \Delta T_{max}$)	33.12	2.43	9.97	0.67	7.40
p-value	<0.001	0.120	0.002	0.415	0.007
Observations			291		
<i>Panel C. Districts — Urban population</i>					
ΔT_{min}	0.010 (2.55)	-0.003 (-0.68)	-0.011 (-2.60)	0.001 (0.08)	-0.002 (-0.33)
ΔT_{max}	0.000 (0.03)	0.002 (0.50)	0.004 (0.74)	-0.006 (-0.51)	0.000 (0.01)
F ($\Delta T_{min} = \Delta T_{max}$)	2.39	0.75	6.07	0.28	0.06
p-value	0.123	0.387	0.014	0.596	0.810
Observations			291		
<i>Panel D. Census towns</i>					
ΔT_{min}	0.021 (5.67)	0.000 (0.01)	-0.004 (-0.92)	0.000 (0.05)	0.019 (2.97)
ΔT_{max}	0.014 (2.65)	0.006 (1.39)	0.008 (1.66)	-0.014 (-1.50)	0.012 (1.13)
F ($\Delta T_{min} = \Delta T_{max}$)	1.55	1.45	4.95	2.30	0.42
p-value	0.213	0.228	0.026	0.130	0.516
Observations			2,969		
Rainfall control	Y	Y	Y	Y	Y
State FE	Y	Y	Y	Y	Y

Notes: Long-difference estimates (1981–2011) of growing-season nighttime (ΔT_{min}) and daytime (ΔT_{max}) warming on per-capita worker rates—each category’s worker count divided by population. The four mutually exclusive categories (columns 1–4) sum to the work-participation rate (column 5). Panels A–C split each district’s population into total, rural, and urban (the within-district residence split); Panel D is the balanced panel of Census towns. All specifications enter T_{min} and T_{max} jointly with a rainfall control and state fixed effects; parentheses report t -statistics from district-clustered standard errors, and coefficients with $|t| > 2$ are in bold. The F -statistic and its p -value test $\Delta T_{min} = \Delta T_{max}$. Compare with the worker-share results in Figure 2 (district rural/urban) and Table 3 (Census towns) of the main text.

Appendix F. Individual vs Simultaneous Temperature Specifications

TABLE A13. ICRISAT Districts, Long Difference: Individual vs Simultaneous Entry

	Cultivated Area (1)	Yield Index (2)	Field Wage (3)	Price Index (4)
<i>T_{wmin} only</i>				
ΔT_{wmin}	-0.391*** (0.1210) [0.1172]	0.109 (0.0953) [0.0923]	-0.136 (0.1054) [0.1062]	-0.064 (0.0491) [0.0505]
Observations	311	303	259	294
<i>T_{wmax} only</i>				
ΔT_{wmax}	-1.218*** (0.3565) [0.3454]	-0.011 (0.2917) [0.2829]	-0.431** (0.1843) [0.1833]	-0.066 (0.0948) [0.0987]
Observations	311	303	259	294
<i>Simultaneous (baseline)</i>				
ΔT_{wmin}	0.289 (0.2017) [0.1954]	0.357** (0.1668) [0.1601]	0.144 (0.1904) [0.1849]	-0.118 (0.1013) [0.0981]
ΔT_{wmax}	-1.715*** (0.6231) [0.6034]	-0.626 (0.5064) [0.4897]	-0.675** (0.3403) [0.3204]	0.138 (0.2002) [0.1943]
F-stat ($\Delta T_{wmin} = \Delta T_{wmax}$)	6.15	2.23	2.57	0.76
Observations	311	303	259	294
<i>Simultaneous, no rainfall control</i>				
ΔT_{wmin}	0.230 (0.1883) [0.1852]	0.343** (0.1570) [0.1517]	0.140 (0.1878) [0.1818]	-0.112 (0.0969) [0.0948]
ΔT_{wmax}	-1.643*** (0.6053) [0.5896]	-0.610 (0.4905) [0.4756]	-0.669** (0.3372) [0.3179]	0.129 (0.1938) [0.1894]
F-stat ($\Delta T_{wmin} = \Delta T_{wmax}$)	5.85	2.29	2.56	0.73
Observations	311	303	259	294
Dep. var. mean	-0.076	0.368	1.919	1.364

Notes: Blocks enter T_{wmin} and T_{wmax} one at a time, simultaneously (the baseline decomposition), and simultaneously without the rainfall control. Given the high correlation between the two measures ($\rho = 0.93$), individual entry illustrates the omitted-variable bias the simultaneous specification removes. Long-differenced cross-section: outcomes and climate averaged over 1981–91 (baseline) and 2001–11 (endline), then differenced. All specifications include state fixed effects. Temperatures are wet-bulb unless noted, constructed as a moving average of the preceding decade’s growing seasons (June–February), as inverse-distance-squared weighted averages of surrounding IMD grid points (see Appendix C.3). All specifications control for total precipitation over the same window unless noted. Standard errors in parentheses, clustered at the district level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

TABLE A14. Census Districts, Long Difference: Individual vs Simultaneous Entry

	Seasonal (1)	Ag. Labour (2)	Cultivation (3)	Non-Agriculture (4)
<i>T_{wmin} only</i>				
ΔT_{wmin}	-0.087*** (0.0250) [0.0238]	0.092*** (0.0201) [0.0195]	-0.022 (0.0290) [0.0279]	0.021 (0.0261) [0.0251]
Observations	293	293	293	293
<i>T_{wmax} only</i>				
ΔT_{wmax}	-0.107** (0.0479) [0.0456]	0.109** (0.0428) [0.0413]	-0.047 (0.0574) [0.0551]	0.050 (0.0539) [0.0521]
Observations	293	293	293	293
<i>Simultaneous (baseline)</i>				
ΔT_{wmin}	-0.147*** (0.0440) [0.0430]	0.159*** (0.0329) [0.0329]	-0.009 (0.0512) [0.0497]	0.001 (0.0469) [0.0451]
ΔT_{wmax}	0.144* (0.0863) [0.0848]	-0.165** (0.0652) [0.0649]	-0.031 (0.1004) [0.0974]	0.049 (0.0968) [0.0936]
F-stat ($\Delta T_{wmin} = \Delta T_{wmax}$)	5.42	11.95	0.02	0.12
Observations	293	293	293	293
<i>Simultaneous, no rainfall control</i>				
ΔT_{wmin}	-0.147*** (0.0442) [0.0433]	0.155*** (0.0333) [0.0333]	-0.009 (0.0506) [0.0492]	0.006 (0.0461) [0.0443]
ΔT_{wmax}	0.145* (0.0863) [0.0849]	-0.161** (0.0671) [0.0670]	-0.031 (0.0996) [0.0968]	0.044 (0.0969) [0.0938]
F-stat ($\Delta T_{wmin} = \Delta T_{wmax}$)	5.44	10.81	0.02	0.08
Observations	293	293	293	293
Dep. var. mean	0.141	-0.049	-0.141	0.049

Notes: Blocks enter T_{wmin} and T_{wmax} one at a time, simultaneously (the baseline decomposition), and simultaneously without the rainfall control. Outcomes are shares of total district workers. Long-differenced cross-section: outcomes and climate averaged over 1981–91 (baseline) and 2001–11 (endline), then differenced. All specifications include state fixed effects. Temperatures are wet-bulb unless noted, constructed as a moving average of the preceding decade's growing seasons (June–February), as inverse-distance-squared weighted averages of surrounding IMD grid points (see Appendix C.3). All specifications control for total precipitation over the same window unless noted. Standard errors in parentheses, clustered at the district level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

TABLE A15. Census Towns, Long Difference: Individual vs Simultaneous Entry

	Seasonal (1)	Ag. Labour (2)	Cultivation (3)	Non-Agriculture (4)
<i>T_{wmin} only</i>				
ΔT_{wmin}	-0.021* (0.0115) [0.0120]	0.050*** (0.0121) [0.0127]	-0.016 (0.0113) [0.0119]	-0.013 (0.0174) [0.0186]
Observations	2,969	2,969	2,969	2,969
<i>T_{wmax} only</i>				
ΔT_{wmax}	-0.017 (0.0213) [0.0221]	0.123*** (0.0213) [0.0229]	0.002 (0.0191) [0.0193]	-0.108*** (0.0330) [0.0348]
Observations	2,969	2,969	2,969	2,969
<i>Simultaneous (baseline)</i>				
ΔT_{wmin}	-0.042** (0.0209) [0.0231]	0.003 (0.0225) [0.0242]	-0.048** (0.0220) [0.0237]	0.087*** (0.0308) [0.0331]
ΔT_{wmax}	0.051 (0.0390) [0.0432]	0.119*** (0.0419) [0.0463]	0.079** (0.0398) [0.0414]	-0.249*** (0.0612) [0.0648]
F-stat ($\Delta T_{wmin} = \Delta T_{wmax}$)	2.63	3.47	4.50	14.29
Observations	2,969	2,969	2,969	2,969
<i>Simultaneous, no rainfall control</i>				
ΔT_{wmin}	-0.049** (0.0218) [0.0241]	0.002 (0.0224) [0.0246]	-0.044** (0.0223) [0.0238]	0.091*** (0.0308) [0.0336]
ΔT_{wmax}	0.052 (0.0407) [0.0450]	0.119*** (0.0419) [0.0463]	0.079* (0.0403) [0.0423]	-0.249*** (0.0613) [0.0650]
F-stat ($\Delta T_{wmin} = \Delta T_{wmax}$)	2.82	3.53	4.06	14.55
Observations	2,969	2,969	2,969	2,969
Dep. var. mean	0.102	-0.053	-0.058	0.010

Notes: Blocks enter T_{wmin} and T_{wmax} one at a time, simultaneously (the baseline decomposition), and simultaneously without the rainfall control. Sample is the balanced panel of towns existing since 1981. Long-differenced cross-section: outcomes and climate averaged over 1981–91 (baseline) and 2001–11 (endline), then differenced. All specifications include state fixed effects. Temperatures are wet-bulb unless noted, constructed as a moving average of the preceding decade's growing seasons (June–February), as inverse-distance-squared weighted averages of surrounding IMD grid points (see Appendix C.3). All specifications control for total precipitation over the same window unless noted. Standard errors in parentheses, clustered at the district level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Appendix G. Further Robustness: Diurnal Basis, Spatial Radius, and Town Leave-One-Out

TABLE A16. Multicollinearity Diagnostics: Surface-Air vs Wet-Bulb

	Surface-air (T_{min}, T_{max})		Wet-bulb (T_{wmin}, T_{wmax})	
	Raw	Within	Raw	Within
Correlation ρ	0.78	-0.09	0.97	0.84
Variance-inflation factor	3.3	1.0	21.6	3.5
Condition number	3.3	1.1	9.2	3.4

Notes: Correlation, variance-inflation factor, and condition number for the temperature pair in the district decomposition—surface-air (T_{min}, T_{max}) and wet-bulb (T_{wmin}, T_{wmax})—raw and after the baseline within transformation (district and decade fixed effects, state linear trend, rainfall included). The surface-air baseline is near-orthogonal in its estimating variation (within VIF near 1); the wet-bulb pair stays collinear within transformation, explaining the fragility of its daytime coefficient.

TABLE A17. Diurnal Mean and Range Decomposition

Panel A. ICRI SAT Districts (1981-2011)					
	Cultivated Area (1)	Production (2)	Harvest Price Index (3)	Field Wage (4)	Yield Index (5)
Diurnal Mean	-0.444*** (0.0848)	-0.366*** (0.0950)	0.088* (0.0453)	-0.157* (0.0898)	0.006 (0.0647)
Diurnal Range	-0.003 (0.0289)	-0.038 (0.0427)	0.086*** (0.0236)	-0.041 (0.0477)	-0.049 (0.0300)
Observations	1,243	1,243	1,191	1,115	1,211
Dep Var. Mean	5.45	5.77	5.66	2.86	7.14
Panel B. Census Districts (1981-2011)					
	Ag. Labour (1)	Cultivator (2)	Seasonal (3)	Non-Agriculture (4)	Log Workers (5)
Diurnal Mean	0.018 (0.0128)	-0.038** (0.0169)	-0.001 (0.0178)	0.021 (0.0171)	0.159 (0.1162)
Diurnal Range	-0.035*** (0.0070)	0.009 (0.0092)	0.032*** (0.0084)	-0.006 (0.0085)	-0.096 (0.0625)
Observations	1,187	1,187	1,187	1,187	1,187
Dep Var. Mean	0.18	0.32	0.17	0.32	13.45
Rainfall Control	Y	Y	Y	Y	Y
State Linear Trend	Y	Y	Y	Y	Y

Notes: Reparameterizes the baseline (T_{min}, T_{max}) decomposition into the growing-season diurnal mean $\frac{1}{2}(T_{min} + T_{max})$ and range ($T_{max} - T_{min}$). The transformation is exact and linear, so the range coefficient equals $\frac{1}{2}(\beta_{T_{max}} - \beta_{T_{min}})$ and its significance is identical to the night/day equality test of the main decomposition. The diurnal range carries the labor reallocation and the harvest-price response; the common cultivated-area contraction loads on the mean. Decadal panel, unit and decade fixed effects and a state $\times$ decade linear trend, precipitation control; district-clustered standard errors.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

TABLE A18. Conley Spatial-HAC Standard Errors: Radius Sweep

		Coefficient	Standard error by estimator			
			Cluster	50 km	100 km	150 km
Panel A. ICRISAT Districts						
Cultivated area						
T_{min}	-0.218***	(0.0459)	(0.0459)	(0.0524)	(0.0578)	(0.0611)
T_{max}	-0.225***	(0.0562)	(0.0560)	(0.0630)	(0.0684)	(0.0689)
Harvest price index						
T_{min}	-0.042	(0.0311)	(0.0311)	(0.0359)	(0.0396)	(0.0446)
T_{max}	0.130***	(0.0341)	(0.0345)	(0.0361)	(0.0392)	(0.0430)
Field wage						
T_{min}	-0.038	(0.0603)	(0.0599)	(0.0689)	(0.0792)	(0.0857)
T_{max}	-0.119*	(0.0703)	(0.0677)	(0.0795)	(0.0898)	(0.0971)
Panel B. Census Districts						
Ag. wage labour share						
T_{min}	0.044***	(0.0096)	(0.0095)	(0.0106)	(0.0115)	(0.0129)
T_{max}	-0.026***	(0.0094)	(0.0094)	(0.0115)	(0.0127)	(0.0141)
Seasonal share						
T_{min}	-0.033***	(0.0115)	(0.0115)	(0.0141)	(0.0165)	(0.0187)
T_{max}	0.032**	(0.0129)	(0.0130)	(0.0150)	(0.0173)	(0.0178)
Non-agriculture share						
T_{min}	0.017	(0.0114)	(0.0112)	(0.0114)	(0.0111)	(0.0102)
T_{max}	0.004	(0.0127)	(0.0125)	(0.0131)	(0.0129)	(0.0129)

Notes: Baseline decadal-panel point estimates; the standard-error columns sweep the Conley spatial-temporal HAC radius from 50 to 200 km (four decadal time lags throughout), spanning the residual spatial-correlation decay distance for districts. Standard errors rise with the radius but the headline night/day labor split and the daytime harvest-price response remain significant at the widest (200 km) radius. Coefficient stars are district-clustered. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

TABLE A19. Peninsular-South Town Result: Leave-One-State-Out

	Ag. Wage Labour (1)	Non- Agriculture (2)	Log Total Workers (3)	Towns
Full peninsular South	0.111*** (0.020)	-0.128*** (0.027)	0.544* (0.290)	1,351
– Odisha	0.130*** (0.023)	-0.152*** (0.032)	0.548 (0.352)	1,246
– Chhattisgarh	0.113*** (0.021)	-0.129*** (0.028)	0.573* (0.307)	1,305
– Madhya Pradesh	0.104*** (0.025)	-0.135*** (0.031)	0.563 (0.374)	1,102
– Maharashtra	0.111*** (0.021)	-0.114*** (0.030)	0.269 (0.195)	1,080
– Andhra Pradesh	0.127*** (0.021)	-0.150*** (0.029)	0.664** (0.331)	1,205
– Karnataka	0.105*** (0.021)	-0.120*** (0.028)	0.509* (0.297)	1,142
– Kerala	0.091*** (0.018)	-0.114*** (0.028)	0.686** (0.285)	1,265
– Tamil Nadu	0.107*** (0.019)	-0.113*** (0.026)	0.495* (0.292)	1,112

Notes: Long-difference (1981–2011) nighttime (T_{min}) coefficient on the balanced peninsular-south Census-town sample, dropping each southern state in turn, to test whether the headline southern result is driven by a single state. The agricultural wage-labor rise and the non-agricultural contraction are stable across every leave-one-out sample (all $p < 0.01$); the total-workforce response is more state-sensitive. District-clustered standard errors. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Appendix H. Additional Controls and Subsamples

TABLE A20. Census Towns, Panel: Baseline-Size-Specific Trends

	Seasonal (1)	Ag. Labour (2)	Cultivation (3)	Non-Agriculture (4)
<i>Baseline</i>				
T_{wmin}	-0.023** (0.0113) [0.0106]	0.030** (0.0118) [0.0100]	-0.017* (0.0094) [0.0084]	0.008 (0.0160) [0.0148]
T_{wmax}	0.046** (0.0184) [0.0177]	-0.025 (0.0190) [0.0165]	0.016 (0.0133) [0.0125]	-0.037 (0.0258) [0.0237]
F-stat ($T_{wmin} = T_{wmax}$)	5.91	3.37	2.15	1.29
Observations	16,215	16,215	16,215	16,215
<i>Adding 1961 population $\times$ decade trend</i>				
T_{wmin}	-0.023** (0.0113) [0.0106]	0.031*** (0.0118) [0.0100]	-0.016* (0.0093) [0.0084]	0.008 (0.0159) [0.0147]
T_{wmax}	0.047** (0.0184) [0.0177]	-0.026 (0.0189) [0.0165]	0.015 (0.0133) [0.0125]	-0.036 (0.0258) [0.0238]
F-stat ($T_{wmin} = T_{wmax}$)	5.99	3.50	1.98	1.19
Observations	16,215	16,215	16,215	16,215
Dep. var. mean	0.095	0.094	0.077	0.732

Notes: The second block adds a predetermined town-size control: 1961 population interacted with a linear decade trend, allowing initially larger towns their own transformation trajectory. Decadal panel (1981–2011) with unit and decade fixed effects and a state $\times$ decade linear trend. Temperatures are wet-bulb unless noted, constructed as a moving average of the preceding decade's growing seasons (June–February), as inverse-distance-squared weighted averages of surrounding IMD grid points (see Appendix C.3). All specifications control for total precipitation over the same window unless noted. Standard errors in parentheses, clustered at the district level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

TABLE A21. ICRISAT Districts, Panel: Baseline Irrigation Subsample

	Cultivated Area (1)	Yield Index (2)	Field Wage (3)	Price Index (4)
<i>All districts</i>				
T_{wmin}	0.090 (0.0781) [0.0778]	0.153** (0.0773) [0.0770]	0.137 (0.1243) [0.1204]	-0.237*** (0.0598) [0.0598]
T_{wmax}	-0.579*** (0.1534) [0.1529]	-0.194 (0.1280) [0.1270]	-0.328* (0.1928) [0.1854]	0.366*** (0.0936) [0.0946]
F-stat ($T_{wmin} = T_{wmax}$)	9.01	3.07	2.28	16.59
Observations	1,243	1,211	1,115	1,191
<i>Above-median baseline irrigation share</i>				
T_{wmin}	0.126 (0.0941) [0.0929]	0.240* (0.1269) [0.1249]	0.382** (0.1830) [0.1747]	-0.177** (0.0810) [0.0797]
T_{wmax}	-0.725*** (0.1720) [0.1709]	-0.281 (0.1810) [0.1785]	-0.762*** (0.2799) [0.2665]	0.348** (0.1428) [0.1419]
F-stat ($T_{wmin} = T_{wmax}$)	11.41	3.11	6.55	5.82
Observations	664	636	586	633
Dep. var. mean	5.451	7.142	2.856	5.660

Notes: The second block restricts the sample to districts with above-median predetermined (Green-Revolution era) irrigation shares. The persistence of the T_{wmin} yield dividend within the irrigated belt indicates nighttime moisture is not a proxy for irrigation. Decadal panel (1981–2011) with unit and decade fixed effects and a state $\times$ decade linear trend. Temperatures are wet-bulb unless noted, constructed as a moving average of the preceding decade's growing seasons (June–February), as inverse-distance-squared weighted averages of surrounding IMD grid points (see Appendix C.3). All specifications control for total precipitation over the same window unless noted. Standard errors in parentheses, clustered at the district level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Appendix I. Total, Rural, and Urban Decomposition in Districts

TABLE A22. Census Districts, Panel: Total vs Rural vs Urban Worker Shares

	Seasonal (Total) (1)	Seasonal (Rural) (2)	Seasonal (Urban) (3)	Non-Ag. (Total) (4)	Non-Ag. (Rural) (5)	Non-Ag. (Urban) (6)
<i>Wet-bulb T_{wmin}/T_{wmax} (baseline)</i>						
T_{wmin}	-0.083*** (0.0216) [0.0214]	-0.087*** (0.0225) [0.0223]	-0.007 (0.0120) [0.0118]	0.011 (0.0221) [0.0217]	0.022 (0.0192) [0.0191]	0.009 (0.0233) [0.0227]
T_{wmax}	0.095*** (0.0349) [0.0352]	0.111*** (0.0373) [0.0373]	-0.011 (0.0193) [0.0192]	0.009 (0.0355) [0.0351]	-0.014 (0.0311) [0.0309]	-0.010 (0.0375) [0.0366]
F-stat ($T_{wmin} = T_{wmax}$)	10.68	11.97	0.01	0.00	0.52	0.11
Observations	1,187	1,177	1,176	1,187	1,177	1,176
<i>Degree days, growing season</i>						
Degree Days	-0.020 (0.0182) [0.0186]	-0.012 (0.0205) [0.0208]	-0.022** (0.0105) [0.0106]	0.025 (0.0178) [0.0175]	0.015 (0.0144) [0.0142]	0.004 (0.0174) [0.0172]
Observations	1,187	1,177	1,176	1,187	1,177	1,176
Dep. var. mean	0.173	0.193	0.080	0.324	0.212	0.802

Notes: Columns decompose district worker shares into total, rural, and urban components for seasonal (marginal) and non-agricultural workers. Decadal panel (1981–2011) with unit and decade fixed effects and a state $\times$ decade linear trend. Temperatures are wet-bulb unless noted, constructed as a moving average of the preceding decade's growing seasons (June–February), as inverse-distance-squared weighted averages of surrounding IMD grid points (see Appendix C.3). All specifications control for total precipitation over the same window unless noted. Standard errors in parentheses, clustered at the district level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

TABLE A23. Census Districts, Long Difference: Total vs Rural vs Urban Worker Shares

	Seasonal (Total) (1)	Seasonal (Rural) (2)	Seasonal (Urban) (3)	Non-Ag. (Total) (4)	Non-Ag. (Rural) (5)	Non-Ag. (Urban) (6)
<i>Wet-bulb T_{wmin}/T_{wmax} (baseline)</i>						
ΔT_{wmin}	-0.147*** (0.0440) [0.0430]	-0.150*** (0.0452) [0.0440]	-0.037 (0.0238) [0.0229]	0.001 (0.0469) [0.0451]	0.010 (0.0372) [0.0363]	0.057 (0.0527) [0.0504]
ΔT_{wmax}	0.144* (0.0863) [0.0848]	0.171* (0.0924) [0.0899]	0.018 (0.0424) [0.0410]	0.049 (0.0968) [0.0936]	-0.007 (0.0745) [0.0724]	-0.104 (0.0988) [0.0945]
F-stat ($\Delta T_{wmin} = \Delta T_{wmax}$)	5.42	5.91	0.76	0.12	0.02	1.21
Observations	293	291	291	293	291	291
<i>Degree days, growing season</i>						
Δ Degree Days	-0.086* (0.0484) [0.0463]	-0.069 (0.0523) [0.0495]	-0.032 (0.0234) [0.0225]	0.053 (0.0529) [0.0512]	0.012 (0.0372) [0.0356]	-0.009 (0.0479) [0.0459]
Observations	293	291	291	293	291	291
Dep. var. mean	0.141	0.154	0.097	0.049	0.043	-0.023

Notes: Columns decompose district worker shares into total, rural, and urban components for seasonal (marginal) and non-agricultural workers. Long-differenced cross-section: outcomes and climate averaged over 1981–91 (baseline) and 2001–11 (endline), then differenced. All specifications include state fixed effects. Temperatures are wet-bulb unless noted, constructed as a moving average of the preceding decade’s growing seasons (June–February), as inverse-distance-squared weighted averages of surrounding IMD grid points (see Appendix C.3). All specifications control for total precipitation over the same window unless noted. Standard errors in parentheses, clustered at the district level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Appendix J. Crop-Level Effects

TABLE A24. ICRISAT Districts, Long Difference: Log Yields by Crop

	Rice (1)	Wheat (2)	Maize (3)	Cereal Index (4)
<i>Wet-bulb T_{wmin}/T_{wmax} (baseline)</i>				
ΔT_{wmin}	0.269 (0.2156) [0.2038]	-0.450* (0.2453) [0.2071]	0.251 (0.2794) [0.2626]	0.357** (0.1668) [0.1601]
ΔT_{wmax}	-0.388 (0.5826) [0.5526]	0.833 (0.5357) [0.4463]	-0.038 (0.5907) [0.5537]	-0.626 (0.5064) [0.4897]
F-stat ($\Delta T_{wmin} = \Delta T_{wmax}$)	0.74	2.79	0.12	2.23
Observations	288	270	272	303
<i>Degree days, growing season</i>				
Δ Degree Days	0.102 (0.3731) [0.3543]	0.173 (0.2234) [0.2007]	0.331 (0.3016) [0.2883]	-0.072 (0.3021) [0.2928]
Observations	288	270	272	303
Dep. var. mean	0.275	0.259	0.387	0.368

Notes: Outcomes are log area-weighted yields by individual crop; the cereal index aggregates rice, wheat, barley, maize, millets, and sorghum weighted by baseline (1966–71) land shares. Long-differenced cross-section: outcomes and climate averaged over 1981–91 (baseline) and 2001–11 (endline), then differenced. All specifications include state fixed effects. Temperatures are wet-bulb unless noted, constructed as a moving average of the preceding decade’s growing seasons (June–February), as inverse-distance-squared weighted averages of surrounding IMD grid points (see Appendix C.3). All specifications control for total precipitation over the same window unless noted. Standard errors in parentheses, clustered at the district level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

TABLE A25. ICRISAT Districts, Long Difference: Log Cultivated Area by Crop

	Rice (1)	Wheat (2)	Maize (3)	Sorghum (4)	Barley (5)	Millet (6)
<i>Growing season (baseline)</i>						
ΔT_{wmin}	1.056* (0.5538) [0.5427]	-0.476 (0.5143) [0.5243]	-2.968*** (0.9474) [0.9843]	-3.699*** (1.0073) [0.9566]	0.138 (1.0694) [1.0711]	-1.524* (0.8694) [0.8436]
ΔT_{wmax}	-1.842 (1.2793) [1.2079]	1.292 (0.9199) [0.9416]	4.086* (2.1837) [2.2175]	3.305 (2.4173) [2.3062]	-0.106 (2.5233) [2.5422]	2.665 (1.7011) [1.6408]
F-stat ($\Delta T_{wmin} = \Delta T_{wmax}$)	2.75	1.60	5.22	4.37	0.00	2.79
Observations	299	281	293	259	178	281
<i>Calendar year</i>						
ΔT_{wmin}	1.274* (0.7085) [0.6867]	-0.949 (0.6698) [0.6799]	-3.070** (1.2015) [1.2231]	-3.845*** (1.3932) [1.3286]	-0.364 (1.4486) [1.4282]	-1.772* (1.0594) [1.0000]
ΔT_{wmax}	-2.015 (1.5236) [1.4383]	2.228* (1.2034) [1.2229]	3.780 (2.4894) [2.4854]	3.255 (2.9287) [2.7993]	0.706 (3.0216) [3.0063]	2.816 (1.9548) [1.8364]
F-stat ($\Delta T_{wmin} = \Delta T_{wmax}$)	2.29	2.97	3.51	2.76	0.06	2.40
Observations	299	281	293	259	178	281
Dep. var. mean	-0.094	-0.082	0.153	-0.911	-1.105	-0.816

Notes: Outcomes are log cultivated area by individual crop; the cereal index aggregates rice, wheat, barley, maize, millets, and sorghum weighted by baseline (1966–71) land shares. Long-differenced cross-section: outcomes and climate averaged over 1981–91 (baseline) and 2001–11 (endline), then differenced. All specifications include state fixed effects. Temperatures are wet-bulb unless noted, constructed as a moving average of the preceding decade’s growing seasons (June–February), as inverse-distance-squared weighted averages of surrounding IMD grid points (see Appendix C.3). All specifications control for total precipitation over the same window unless noted. Standard errors in parentheses, clustered at the district level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

TABLE A26. ICRISAT Districts, Long Difference: Log Harvest Prices by Crop

	Rice (1)	Wheat (2)	Maize (3)	Sorghum (4)	Barley (5)	Millet (6)
<i>Growing season (baseline)</i>						
ΔT_{wmin}	-0.050 (0.1817) [0.1762]	0.236** (0.1085) [0.1094]	0.202 (0.1918) [0.1787]	0.173 (0.2129) [0.2027]	0.454** (0.1863) [0.1758]	-0.840 (0.5386) [0.5461]
ΔT_{wmax}	0.003 (0.2643) [0.2504]	-0.364* (0.1968) [0.2004]	-0.204 (0.3740) [0.3401]	-0.122 (0.4291) [0.4079]	-1.428** (0.5562) [0.5325]	1.231 (0.8736) [0.8949]
F-stat ($\Delta T_{wmin} = \Delta T_{wmax}$)	0.02	4.00	0.54	0.22	6.57	2.23
Observations	271	236	217	210	132	210
<i>Calendar year</i>						
ΔT_{wmin}	-0.079 (0.1970) [0.1880]	0.188 (0.1173) [0.1173]	0.203 (0.2078) [0.1950]	0.046 (0.2801) [0.2665]	0.702** (0.3025) [0.3127]	-0.814 (0.5652) [0.5741]
ΔT_{wmax}	0.060 (0.2884) [0.2696]	-0.198 (0.2007) [0.1999]	-0.216 (0.3797) [0.3392]	0.179 (0.5197) [0.4987]	-1.663** (0.7104) [0.7531]	0.947 (0.9271) [0.9218]
F-stat ($\Delta T_{wmin} = \Delta T_{wmax}$)	0.09	1.53	0.53	0.03	5.50	1.45
Observations	271	236	217	210	132	210
Dep. var. mean	1.369	1.369	1.378	1.424	1.444	1.343

Notes: Outcomes are log harvest prices by individual crop; the cereal index aggregates rice, wheat, barley, maize, millets, and sorghum weighted by baseline (1966–71) land shares. Long-differenced cross-section: outcomes and climate averaged over 1981–91 (baseline) and 2001–11 (endline), then differenced. All specifications include state fixed effects. Temperatures are wet-bulb unless noted, constructed as a moving average of the preceding decade’s growing seasons (June–February), as inverse-distance-squared weighted averages of surrounding IMD grid points (see Appendix C.3). All specifications control for total precipitation over the same window unless noted. Standard errors in parentheses, clustered at the district level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Appendix K. Degree-Days Exposure

TABLE A27. Census Districts, Long Difference: Degree Days by Season

	Seasonal (1)	Ag. Labour (2)	Cultivation (3)	Non-Agriculture (4)
<i>Growing season (baseline)</i>				
Δ Degree Days	-0.086* (0.0484) [0.0463]	0.080** (0.0402) [0.0390]	-0.043 (0.0565) [0.0542]	0.053 (0.0529) [0.0512]
Observations	293	293	293	293
<i>Kharif</i>				
Δ Degree Days	-0.045 (0.0425) [0.0408]	0.105** (0.0416) [0.0401]	-0.072 (0.0557) [0.0537]	0.020 (0.0550) [0.0533]
Observations	293	293	293	293
<i>Rabi</i>				
Δ Degree Days	-0.088** (0.0437) [0.0419]	0.026 (0.0338) [0.0326]	-0.004 (0.0488) [0.0466]	0.066 (0.0423) [0.0405]
Observations	293	293	293	293
<i>Lean</i>				
Δ Degree Days	-0.146*** (0.0409) [0.0391]	0.110*** (0.0344) [0.0334]	-0.028 (0.0519) [0.0500]	0.068 (0.0426) [0.0404]
Observations	293	293	293	293
<i>Calendar year</i>				
Δ Degree Days	-0.123** (0.0503) [0.0478]	0.111*** (0.0410) [0.0396]	-0.047 (0.0583) [0.0560]	0.063 (0.0559) [0.0539]
Observations	293	293	293	293
Dep. var. mean	0.141	-0.049	-0.141	0.049

Notes: Blocks replace the wet-bulb min/max decomposition with growing-degree-days (bounded 8–32°C) over alternative seasons. Long-differenced cross-section: outcomes and climate averaged over 1981–91 (baseline) and 2001–11 (endline), then differenced. All specifications include state fixed effects. Temperatures are wet-bulb unless noted, constructed as a moving average of the preceding decade's growing seasons (June–February), as inverse-distance-squared weighted averages of surrounding IMD grid points (see Appendix C.3). All specifications control for total precipitation over the same window unless noted. Standard errors in parentheses, clustered at the district level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

TABLE A28. Census Districts, Panel: Degree Days by Season

	Seasonal (1)	Ag. Labour (2)	Cultivation (3)	Non-Agriculture (4)
<i>Growing season (baseline)</i>				
Degree Days	-0.020 (0.0182) [0.0186]	0.036*** (0.0138) [0.0136]	-0.041** (0.0178) [0.0176]	0.025 (0.0178) [0.0175]
Observations	1,187	1,187	1,187	1,187
<i>Kharif</i>				
Degree Days	0.032** (0.0138) [0.0139]	-0.009 (0.0108) [0.0107]	-0.036** (0.0141) [0.0139]	0.015 (0.0138) [0.0136]
Observations	1,187	1,187	1,187	1,187
<i>Rabi</i>				
Degree Days	-0.045*** (0.0119) [0.0123]	0.043*** (0.0090) [0.0089]	-0.016 (0.0118) [0.0118]	0.018 (0.0108) [0.0106]
Observations	1,187	1,187	1,187	1,187
<i>Lean</i>				
Degree Days	-0.008 (0.0088) [0.0089]	0.018*** (0.0066) [0.0065]	-0.029*** (0.0085) [0.0087]	0.018** (0.0080) [0.0078]
Observations	1,187	1,187	1,187	1,187
<i>Calendar year</i>				
Degree Days	-0.011 (0.0151) [0.0154]	0.033*** (0.0117) [0.0115]	-0.050*** (0.0148) [0.0147]	0.026* (0.0151) [0.0149]
Observations	1,187	1,187	1,187	1,187
Dep. var. mean	0.173	0.180	0.323	0.324

Notes: Blocks replace the wet-bulb min/max decomposition with growing-degree-days (bounded 8–32°C) over alternative seasons. Decadal panel (1981–2011) with unit and decade fixed effects and a state $\times$ decade linear trend. Temperatures are wet-bulb unless noted, constructed as a moving average of the preceding decade's growing seasons (June–February), as inverse-distance-squared weighted averages of surrounding IMD grid points (see Appendix C.3). All specifications control for total precipitation over the same window unless noted. Standard errors in parentheses, clustered at the district level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

TABLE A29. ICRISAT Districts, Long Difference: Degree Days by Season

	Cultivated Area (1)	Yield Index (2)	Field Wage (3)	Price Index (4)
<i>Growing season (baseline)</i>				
ΔDegree Days	-1.133*** (0.3658) [0.3553]	-0.072 (0.3021) [0.2928]	-0.410** (0.1718) [0.1705]	-0.033 (0.0951) [0.0979]
Observations	311	303	259	294
<i>Kharif</i>				
ΔDegree Days	-0.719** (0.3522) [0.3425]	-0.165 (0.2843) [0.2746]	-0.348* (0.1804) [0.1808]	-0.154* (0.0921) [0.0923]
Observations	311	303	259	294
<i>Rabi</i>				
ΔDegree Days	-1.009*** (0.2641) [0.2576]	0.010 (0.2244) [0.2173]	-0.309** (0.1407) [0.1339]	0.057 (0.0875) [0.0890]
Observations	311	303	259	294
<i>Lean</i>				
ΔDegree Days	-0.540* (0.2828) [0.2756]	0.004 (0.2256) [0.2181]	-0.140 (0.1471) [0.1454]	0.028 (0.0908) [0.0912]
Observations	311	303	259	294
<i>Calendar year</i>				
ΔDegree Days	-1.097*** (0.4001) [0.3880]	-0.080 (0.3278) [0.3174]	-0.362** (0.1764) [0.1764]	-0.021 (0.0994) [0.1034]
Observations	311	303	259	294
Dep. var. mean	-0.076	0.368	1.919	1.364

Notes: Blocks replace the wet-bulb min/max decomposition with growing-degree-days (bounded 8–32°C) over alternative seasons. Long-differenced cross-section: outcomes and climate averaged over 1981–91 (baseline) and 2001–11 (endline), then differenced. All specifications include state fixed effects. Temperatures are wet-bulb unless noted, constructed as a moving average of the preceding decade's growing seasons (June–February), as inverse-distance-squared weighted averages of surrounding IMD grid points (see Appendix C.3). All specifications control for total precipitation over the same window unless noted. Standard errors in parentheses, clustered at the district level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

TABLE A30. ICRISAT Districts, Panel: Degree Days by Season

	Cultivated Area (1)	Yield Index (2)	Field Wage (3)	Price Index (4)
<i>Growing season (baseline)</i>				
Degree Days	-0.427*** (0.0872) [0.0871]	0.014 (0.0662) [0.0656]	-0.108 (0.0912) [0.0895]	0.064 (0.0476) [0.0486]
Observations	1,243	1,211	1,115	1,191
<i>Kharif</i>				
Degree Days	-0.236*** (0.0703) [0.0697]	-0.026 (0.0605) [0.0600]	-0.073 (0.0686) [0.0699]	0.008 (0.0408) [0.0413]
Observations	1,243	1,211	1,115	1,191
<i>Rabi</i>				
Degree Days	-0.261*** (0.0531) [0.0540]	0.041 (0.0400) [0.0396]	-0.051 (0.0697) [0.0691]	0.055* (0.0311) [0.0320]
Observations	1,243	1,211	1,115	1,191
<i>Lean</i>				
Degree Days	-0.132*** (0.0430) [0.0453]	-0.027 (0.0266) [0.0263]	0.023 (0.0442) [0.0448]	-0.006 (0.0250) [0.0251]
Observations	1,243	1,211	1,115	1,191
<i>Calendar year</i>				
Degree Days	-0.359*** (0.0817) [0.0834]	0.002 (0.0566) [0.0558]	-0.102 (0.0740) [0.0737]	0.035 (0.0437) [0.0442]
Observations	1,243	1,211	1,115	1,191
Dep. var. mean	5.451	7.142	2.856	5.660

Notes: Blocks replace the wet-bulb min/max decomposition with growing-degree-days (bounded 8–32°C) over alternative seasons. Decadal panel (1981–2011) with unit and decade fixed effects and a state $\times$ decade linear trend. Temperatures are wet-bulb unless noted, constructed as a moving average of the preceding decade's growing seasons (June–February), as inverse-distance-squared weighted averages of surrounding IMD grid points (see Appendix C.3). All specifications control for total precipitation over the same window unless noted. Standard errors in parentheses, clustered at the district level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

TABLE A31. NSS Urban Households, Panel: Degree Days by Season

	Agriculture (1)	Constr. & Mining (2)	Manufacturing (3)	Services (4)
<i>Growing season (baseline)</i>				
Degree Days	-0.019 (0.0545) [0.0522]	-0.013 (0.0399) [0.0385]	0.029 (0.0473) [0.0460]	0.004 (0.0660) [0.0640]
Observations	1,163	1,163	1,163	1,163
<i>Kharif</i>				
Degree Days	0.009 (0.0510) [0.0481]	0.019 (0.0364) [0.0348]	-0.001 (0.0408) [0.0394]	-0.027 (0.0611) [0.0588]
Observations	1,163	1,163	1,163	1,163
<i>Rabi</i>				
Degree Days	-0.033 (0.0407) [0.0395]	-0.027 (0.0358) [0.0349]	0.032 (0.0401) [0.0394]	0.027 (0.0517) [0.0505]
Observations	1,163	1,163	1,163	1,163
<i>Lean</i>				
Degree Days	-0.026 (0.0311) [0.0283]	-0.011 (0.0199) [0.0187]	0.029 (0.0279) [0.0277]	0.008 (0.0380) [0.0368]
Observations	1,163	1,163	1,163	1,163
<i>Calendar year</i>				
Degree Days	-0.037 (0.0559) [0.0526]	-0.017 (0.0384) [0.0367]	0.034 (0.0478) [0.0467]	0.020 (0.0671) [0.0648]
Observations	1,163	1,163	1,163	1,163
Dep. var. mean	0.112	0.094	0.170	0.624

Notes: Blocks replace the wet-bulb min/max decomposition with growing-degree-days (bounded 8–32°C) over alternative seasons. Decadal panel (1981–2011) with unit and decade fixed effects and a state $\times$ decade linear trend. Temperatures are wet-bulb unless noted, constructed as a moving average of the preceding decade's growing seasons (June–February), as inverse-distance-squared weighted averages of surrounding IMD grid points (see Appendix C.3). All specifications control for total precipitation over the same window unless noted. Standard errors in parentheses, clustered at the district level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

TABLE A32. Census Towns, Long Difference: Degree Days by Season

	Seasonal (1)	Ag. Labour (2)	Cultivation (3)	Non-Agriculture (4)
<i>Growing season (baseline)</i>				
Δ Degree Days	-0.005 (0.0210) [0.0225]	0.116*** (0.0208) [0.0229]	-0.002 (0.0183) [0.0186]	-0.110*** (0.0317) [0.0339]
Observations	2,969	2,969	2,969	2,969
<i>Kharif</i>				
Δ Degree Days	-0.000 (0.0205) [0.0215]	0.119*** (0.0218) [0.0230]	-0.024 (0.0186) [0.0187]	-0.093*** (0.0319) [0.0338]
Observations	2,969	2,969	2,969	2,969
<i>Rabi</i>				
Δ Degree Days	-0.019 (0.0170) [0.0186]	0.069*** (0.0173) [0.0184]	0.021 (0.0153) [0.0158]	-0.072*** (0.0250) [0.0263]
Observations	2,969	2,969	2,969	2,969
<i>Lean</i>				
Δ Degree Days	-0.052*** (0.0185) [0.0207]	0.050** (0.0217) [0.0251]	-0.019 (0.0185) [0.0207]	0.021 (0.0293) [0.0341]
Observations	2,969	2,969	2,969	2,969
<i>Calendar year</i>				
Δ Degree Days	-0.020 (0.0216) [0.0231]	0.118*** (0.0215) [0.0238]	-0.014 (0.0195) [0.0200]	-0.084** (0.0333) [0.0368]
Observations	2,969	2,969	2,969	2,969
Dep. var. mean	0.102	-0.053	-0.058	0.010

Notes: Blocks replace the wet-bulb min/max decomposition with growing-degree-days (bounded 8–32°C) over alternative seasons. Long-differenced cross-section: outcomes and climate averaged over 1981–91 (baseline) and 2001–11 (endline), then differenced. All specifications include state fixed effects. Temperatures are wet-bulb unless noted, constructed as a moving average of the preceding decade's growing seasons (June–February), as inverse-distance-squared weighted averages of surrounding IMD grid points (see Appendix C.3). All specifications control for total precipitation over the same window unless noted. Standard errors in parentheses, clustered at the district level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

TABLE A33. Census Towns, Panel: Degree Days by Season

	Seasonal (1)	Ag. Labour (2)	Cultivation (3)	Non-Agriculture (4)
<i>Growing season (baseline)</i>				
Degree Days	0.016 (0.0100) [0.0097]	0.016* (0.0082) [0.0081]	-0.008 (0.0054) [0.0063]	-0.025* (0.0131) [0.0126]
Observations	16,215	16,215	16,215	16,215
<i>Kharif</i>				
Degree Days	0.026*** (0.0079) [0.0078]	-0.008 (0.0063) [0.0066]	-0.016*** (0.0048) [0.0052]	-0.003 (0.0100) [0.0101]
Observations	16,215	16,215	16,215	16,215
<i>Rabi</i>				
Degree Days	-0.003 (0.0068) [0.0066]	0.022*** (0.0058) [0.0055]	0.004 (0.0037) [0.0043]	-0.023*** (0.0089) [0.0086]
Observations	16,215	16,215	16,215	16,215
<i>Lean</i>				
Degree Days	0.000 (0.0048) [0.0050]	0.008* (0.0044) [0.0043]	-0.001 (0.0025) [0.0031]	-0.008 (0.0074) [0.0069]
Observations	16,215	16,215	16,215	16,215
<i>Calendar year</i>				
Degree Days	0.009 (0.0088) [0.0088]	0.012 (0.0084) [0.0079]	-0.005 (0.0048) [0.0056]	-0.016 (0.0132) [0.0122]
Observations	16,215	16,215	16,215	16,215
Dep. var. mean	0.095	0.094	0.077	0.732

Notes: Blocks replace the wet-bulb min/max decomposition with growing-degree-days (bounded 8–32°C) over alternative seasons. Decadal panel (1981–2011) with unit and decade fixed effects and a state $\times$ decade linear trend. Temperatures are wet-bulb unless noted, constructed as a moving average of the preceding decade's growing seasons (June–February), as inverse-distance-squared weighted averages of surrounding IMD grid points (see Appendix C.3). All specifications control for total precipitation over the same window unless noted. Standard errors in parentheses, clustered at the district level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Appendix L. Fixed-Effects and Trend Structure

TABLE A34. Census Districts, Panel: Fixed-Effects Structure

	Seasonal (1)	Ag. Labour (2)	Cultivation (3)	Non-Agriculture (4)
<i>Unit + year + state trend (baseline)</i>				
T_{wmin}	-0.083*** (0.0216) [0.0214]	0.086*** (0.0173) [0.0171]	-0.013 (0.0237) [0.0235]	0.011 (0.0221) [0.0217]
T_{wmax}	0.095*** (0.0349) [0.0352]	-0.079*** (0.0253) [0.0252]	-0.028 (0.0348) [0.0344]	0.009 (0.0355) [0.0351]
F-stat ($T_{wmin} = T_{wmax}$)	10.68	16.27	0.08	0.00
Observations	1,187	1,187	1,187	1,187
<i>Unit + year (two-way)</i>				
T_{wmin}	0.126*** (0.0210) [0.0216]	0.043*** (0.0157) [0.0162]	-0.165*** (0.0226) [0.0234]	-0.006 (0.0160) [0.0158]
T_{wmax}	-0.310*** (0.0343) [0.0359]	0.014 (0.0251) [0.0263]	0.274*** (0.0366) [0.0388]	0.027 (0.0272) [0.0273]
F-stat ($T_{wmin} = T_{wmax}$)	70.34	0.57	59.65	0.67
Observations	1,187	1,187	1,187	1,187
<i>Unit + year + district trend</i>				
T_{wmin}	-0.012 (0.0266) [0.0228]	0.018 (0.0188) [0.0163]	0.005 (0.0205) [0.0176]	-0.010 (0.0168) [0.0145]
T_{wmax}	0.067* (0.0389) [0.0332]	-0.011 (0.0276) [0.0238]	-0.071** (0.0295) [0.0251]	0.014 (0.0280) [0.0241]
F-stat ($T_{wmin} = T_{wmax}$)	1.63	0.42	2.52	0.33
Observations	1,187	1,187	1,187	1,187
<i>Unit + state × year</i>				
T_{wmin}	-0.100*** (0.0291) [0.0284]	0.115*** (0.0226) [0.0220]	-0.014 (0.0327) [0.0316]	0.002 (0.0294) [0.0284]
T_{wmax}	0.093* (0.0479) [0.0476]	-0.130*** (0.0354) [0.0347]	0.002 (0.0514) [0.0499]	0.031 (0.0479) [0.0462]
F-stat ($T_{wmin} = T_{wmax}$)	6.76	19.15	0.04	0.15
Observations	1,164	1,164	1,164	1,164
Dep. var. mean	0.173	0.180	0.323	0.324

Notes: Blocks vary the absorbed fixed effects and trends: unit+year+state trend (baseline), two-way unit+year, unit+year+district trend, and unit+state×year. Decadal panel (1981–2011) with unit and decade fixed effects and a state × decade linear trend. Temperatures are wet-bulb unless noted, constructed as a moving average of the preceding decade's growing seasons (June–February), as inverse-distance-squared weighted averages of surrounding IMD grid points (see Appendix C.3). All specifications control for total precipitation over the same window unless noted. Standard errors in parentheses, clustered at the district level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

TABLE A35. ICRISAT Districts, Panel: Fixed-Effects Structure

	Cultivated Area (1)	Yield Index (2)	Field Wage (3)	Price Index (4)
<i>Unit + year + state trend (baseline)</i>				
T_{wmin}	0.090 (0.0781) [0.0778]	0.153** (0.0773) [0.0770]	0.137 (0.1243) [0.1204]	-0.237*** (0.0598) [0.0598]
T_{wmax}	-0.579*** (0.1534) [0.1529]	-0.194 (0.1280) [0.1270]	-0.328* (0.1928) [0.1854]	0.366*** (0.0936) [0.0946]
F-stat ($T_{wmin} = T_{wmax}$)	9.01	3.07	2.28	16.59
Observations	1,243	1,211	1,115	1,191
<i>Unit + year (two-way)</i>				
T_{wmin}	0.709*** (0.0943) [0.0996]	0.415*** (0.0551) [0.0568]	-0.088 (0.0803) [0.0756]	0.130 (0.0799) [0.0797]
T_{wmax}	-1.239*** (0.1703) [0.1788]	-0.523*** (0.1046) [0.1076]	0.058 (0.1399) [0.1426]	-0.081 (0.1420) [0.1427]
F-stat ($T_{wmin} = T_{wmax}$)	56.99	37.55	0.48	0.95
Observations	1,243	1,211	1,115	1,191
<i>Unit + year + district trend</i>				
T_{wmin}	0.044 (0.0742) [0.0651]	-0.032 (0.0984) [0.0862]	0.169 (0.2244) [0.1716]	-0.288*** (0.0726) [0.0626]
T_{wmax}	-0.250** (0.1142) [0.1031]	0.003 (0.1409) [0.1234]	-0.359 (0.2995) [0.2322]	0.449*** (0.1117) [0.0964]
F-stat ($T_{wmin} = T_{wmax}$)	2.78	0.02	1.07	17.58
Observations	1,243	1,211	1,115	1,191
<i>Unit + state $\times$ year</i>				
T_{wmin}	0.166 (0.1143) [0.1112]	0.273*** (0.1012) [0.0973]	0.063 (0.1354) [0.1242]	-0.086 (0.0706) [0.0690]
T_{wmax}	-0.806*** (0.2418) [0.2343]	-0.313 (0.1914) [0.1845]	-0.415* (0.2136) [0.1969]	0.088 (0.1136) [0.1115]
F-stat ($T_{wmin} = T_{wmax}$)	7.86	4.26	2.02	0.95
Observations	1,243	1,211	1,114	1,190
Dep. var. mean	5.451	7.142	2.856	5.660

Notes: Blocks vary the absorbed fixed effects and trends: unit+year+state trend (baseline), two-way unit+year, unit+year+district trend, and unit+state $\times$ year. Decadal panel (1981–2011) with unit and decade fixed effects and a state $\times$ decade linear trend. Temperatures are wet-bulb unless noted, constructed as a moving average of the preceding decade's growing seasons (June–February), as inverse-distance-squared weighted averages of surrounding IMD grid points (see Appendix C.3). All specifications control for total precipitation over the same window unless noted. Standard errors in parentheses, clustered at the district level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

TABLE A36. NSS Urban Households, Panel: Fixed-Effects Structure

	Agriculture (1)	Constr. & Mining (2)	Manufacturing (3)	Services (4)
<i>Unit + year + state trend (baseline)</i>				
T_{wmin}	0.160*** (0.0614) [0.0600]	0.060 (0.0408) [0.0395]	-0.064 (0.0485) [0.0467]	-0.155** (0.0725) [0.0698]
T_{wmax}	-0.196* (0.1092) [0.1052]	-0.105 (0.0799) [0.0770]	0.091 (0.0880) [0.0846]	0.210 (0.1321) [0.1281]
F-stat ($T_{wmin} = T_{wmax}$)	4.63	1.99	1.41	3.42
Observations	1,163	1,163	1,163	1,163
<i>Unit + year (two-way)</i>				
T_{wmin}	0.077** (0.0383) [0.0388]	0.033 (0.0270) [0.0275]	-0.010 (0.0348) [0.0344]	-0.100** (0.0478) [0.0473]
T_{wmax}	-0.075 (0.0823) [0.0834]	-0.119** (0.0527) [0.0524]	0.035 (0.0627) [0.0625]	0.160* (0.0961) [0.0969]
F-stat ($T_{wmin} = T_{wmax}$)	1.66	3.86	0.22	3.45
Observations	1,163	1,163	1,163	1,163
<i>Unit + state $\times$ year</i>				
T_{wmin}	0.101 (0.0729) [0.0698]	0.046 (0.0500) [0.0478]	0.006 (0.0559) [0.0524]	-0.152* (0.0847) [0.0797]
T_{wmax}	-0.082 (0.1380) [0.1323]	-0.108 (0.1049) [0.1002]	-0.044 (0.1153) [0.1081]	0.233 (0.1672) [0.1587]
F-stat ($T_{wmin} = T_{wmax}$)	0.79	1.03	0.09	2.49
Observations	1,151	1,151	1,151	1,151
Dep. var. mean	0.112	0.094	0.170	0.624

Notes: Blocks vary the absorbed fixed effects and trends: unit+year+state trend (baseline), two-way unit+year, unit+year+district trend, and unit+state $\times$ year. Decadal panel (1981–2011) with unit and decade fixed effects and a state $\times$ decade linear trend. Temperatures are wet-bulb unless noted, constructed as a moving average of the preceding decade's growing seasons (June–February), as inverse-distance-squared weighted averages of surrounding IMD grid points (see Appendix C.3). All specifications control for total precipitation over the same window unless noted. Standard errors in parentheses, clustered at the district level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

TABLE A37. Census Towns, Panel: Fixed-Effects Structure

	Seasonal (1)	Ag. Labour (2)	Cultivation (3)	Non-Agriculture (4)
<i>Unit + year + state trend (baseline)</i>				
T_{wmin}	-0.023** (0.0113) [0.0106]	0.030** (0.0118) [0.0100]	-0.017* (0.0094) [0.0084]	0.008 (0.0160) [0.0148]
T_{wmax}	0.046** (0.0184) [0.0177]	-0.025 (0.0190) [0.0165]	0.016 (0.0133) [0.0125]	-0.037 (0.0258) [0.0237]
F-stat ($T_{wmin} = T_{wmax}$)	5.91	3.37	2.15	1.29
Observations	16,215	16,215	16,215	16,215
<i>Unit + year (two-way)</i>				
T_{wmin}	0.093*** (0.0112) [0.0096]	0.043*** (0.0089) [0.0067]	-0.042*** (0.0104) [0.0088]	-0.092*** (0.0146) [0.0114]
T_{wmax}	-0.165*** (0.0195) [0.0180]	-0.015 (0.0153) [0.0123]	0.056*** (0.0174) [0.0155]	0.122*** (0.0277) [0.0220]
F-stat ($T_{wmin} = T_{wmax}$)	74.53	6.17	13.02	27.40
Observations	16,215	16,215	16,215	16,215
<i>Unit + year + district trend</i>				
T_{wmin}	0.008 (0.0135) [0.0118]	0.024** (0.0112) [0.0094]	0.018** (0.0072) [0.0064]	-0.052*** (0.0161) [0.0140]
T_{wmax}	0.035* (0.0202) [0.0184]	-0.042** (0.0188) [0.0155]	-0.035*** (0.0110) [0.0095]	0.044 (0.0266) [0.0224]
F-stat ($T_{wmin} = T_{wmax}$)	0.73	5.35	9.08	5.36
Observations	16,215	16,215	16,215	16,215
<i>Unit + state $\times$ year</i>				
T_{wmin}	-0.048*** (0.0135) [0.0121]	0.025* (0.0139) [0.0107]	-0.018 (0.0127) [0.0107]	0.043** (0.0177) [0.0161]
T_{wmax}	0.068*** (0.0231) [0.0215]	0.000 (0.0213) [0.0175]	0.025 (0.0185) [0.0162]	-0.100*** (0.0295) [0.0277]
F-stat ($T_{wmin} = T_{wmax}$)	10.54	0.48	2.03	9.94
Observations	16,215	16,215	16,215	16,215
Dep. var. mean	0.095	0.094	0.077	0.732

Notes: Blocks vary the absorbed fixed effects and trends: unit+year+state trend (baseline), two-way unit+year, unit+year+district trend, and unit+state $\times$ year. Decadal panel (1981–2011) with unit and decade fixed effects and a state $\times$ decade linear trend. Temperatures are wet-bulb unless noted, constructed as a moving average of the preceding decade's growing seasons (June–February), as inverse-distance-squared weighted averages of surrounding IMD grid points (see Appendix C.3). All specifications control for total precipitation over the same window unless noted. Standard errors in parentheses, clustered at the district level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Appendix M. Heterogeneity by Baseline Climate and Structure

TABLE A38. Census Districts, Long Difference: Heterogeneity

	Seasonal (1)	Ag. Labour (2)	Cultivation (3)	Non-Agriculture (4)
<i>All units</i>				
ΔT_{wmin}	-0.147*** (0.0440) [0.0430]	0.159*** (0.0329) [0.0329]	-0.009 (0.0512) [0.0497]	0.001 (0.0469) [0.0451]
ΔT_{wmax}	0.144* (0.0863) [0.0848]	-0.165** (0.0652) [0.0649]	-0.031 (0.1004) [0.0974]	0.049 (0.0968) [0.0936]
F-stat ($\Delta T_{wmin} = \Delta T_{wmax}$)	5.42	11.95	0.02	0.12
Observations	293	293	293	293
<i>Hot (high baseline T_{wmin})</i>				
ΔT_{wmin}	-0.002 (0.0819) [0.0773]	0.159*** (0.0584) [0.0553]	-0.114 (0.0821) [0.0777]	-0.050 (0.1035) [0.0982]
ΔT_{wmax}	-0.095 (0.1352) [0.1276]	-0.192* (0.0972) [0.0921]	0.112 (0.1501) [0.1420]	0.186 (0.1782) [0.1691]
F-stat ($\Delta T_{wmin} = \Delta T_{wmax}$)	0.19	5.50	1.01	0.74
Observations	143	143	143	143
<i>Cool (low T_{wmin})</i>				
ΔT_{wmin}	-0.186*** (0.0526) [0.0519]	0.153*** (0.0404) [0.0402]	-0.016 (0.0608) [0.0589]	0.052 (0.0511) [0.0488]
ΔT_{wmax}	0.323** (0.1470) [0.1424]	-0.206* (0.1069) [0.1042]	0.027 (0.1538) [0.1482]	-0.134 (0.1112) [0.1067]
F-stat ($\Delta T_{wmin} = \Delta T_{wmax}$)	7.49	6.51	0.05	1.57
Observations	144	144	144	144
<i>Wet (high baseline rain)</i>				
ΔT_{wmin}	-0.241** (0.1004) [0.0934]	0.154* (0.0780) [0.0739]	0.035 (0.1307) [0.1222]	0.057 (0.0998) [0.0939]
ΔT_{wmax}	0.271 (0.1659) [0.1541]	-0.155 (0.1452) [0.1389]	-0.088 (0.2365) [0.2216]	-0.042 (0.1814) [0.1724]
F-stat ($\Delta T_{wmin} = \Delta T_{wmax}$)	3.81	2.01	0.12	0.13
Observations	144	144	144	144
<i>Dry (low rain)</i>				
ΔT_{wmin}	-0.084* (0.0459) [0.0445]	0.135*** (0.0354) [0.0350]	-0.061 (0.0524) [0.0502]	0.014 (0.0500) [0.0475]
ΔT_{wmax}	0.124 (0.1057) [0.1023]	-0.196** (0.0754) [0.0745]	0.113 (0.1066) [0.1021]	-0.035 (0.1016) [0.0964]
F-stat ($\Delta T_{wmin} = \Delta T_{wmax}$)	2.15	9.71	1.46	0.12
Observations	142	142	142	142
<i>North (high latitude)</i>				
ΔT_{wmin}	-0.182*** (0.0511) [0.0504]	0.159*** (0.0417) [0.0412]	0.000 (0.0582) [0.0561]	0.034 (0.0513) [0.0486]
ΔT_{wmax}	0.167 (0.1255) [0.1230]	-0.088 (0.1015) [0.0991]	0.008 (0.1364) [0.1307]	-0.097 (0.1153) [0.1095]
F-stat ($\Delta T_{wmin} = \Delta T_{wmax}$)	4.49	3.33	0.00	0.71
Observations	148	148	148	148
<i>South (low latitude)</i>				
ΔT_{wmin}	-0.110 (0.0820) [0.0777]	0.193*** (0.0670) [0.0638]	-0.108 (0.0798) [0.0759]	0.012 (0.0958) [0.0912]
ΔT_{wmax}	0.099 (0.1400) [0.1327]	-0.271** (0.1065) [0.1015]	0.116 (0.1454) [0.1382]	0.077 (0.1660) [0.1582]
F-stat ($\Delta T_{wmin} = \Delta T_{wmax}$)	0.92	7.65	1.06	0.07
Observations	145	145	145	145
Dep. var. mean	0.141	-0.049	-0.141	0.049

Notes: Blocks restrict the estimation sample by baseline (long-run) wet-bulb temperature, baseline rainfall, latitude, and, where available, baseline irrigation or 1961 urban status. Long-differenced cross-section: outcomes and climate averaged over 1981–91 (baseline) and 2001–11 (endline), then differenced. All specifications include state fixed effects. Temperatures are wet-bulb unless noted, constructed as a moving average of the preceding decade’s growing seasons (June–February), as inverse-distance-squared weighted averages of surrounding IMD grid points (see Appendix C.3). All specifications control for total precipitation over the same window unless noted. Standard errors in parentheses, clustered at the district level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

TABLE A39. Census Districts, Panel: Heterogeneity

	Seasonal (1)	Ag. Labour (2)	Cultivation (3)	Non-Agriculture (4)
<i>All units</i>				
T_{wmin}	-0.083*** (0.0216) [0.0214]	0.086*** (0.0173) [0.0171]	-0.013 (0.0237) [0.0235]	0.011 (0.0221) [0.0217]
T_{wmax}	0.095*** (0.0349) [0.0352]	-0.079*** (0.0253) [0.0252]	-0.028 (0.0348) [0.0344]	0.009 (0.0355) [0.0351]
F-stat ($T_{wmin} = T_{wmax}$)	10.68	16.27	0.08	0.00
Observations	1,187	1,187	1,187	1,187
<i>Hot (high baseline T_{wmin})</i>				
T_{wmin}	-0.010 (0.0356) [0.0348]	0.018 (0.0264) [0.0256]	-0.013 (0.0317) [0.0310]	-0.001 (0.0356) [0.0348]
T_{wmax}	-0.012 (0.0540) [0.0525]	-0.023 (0.0391) [0.0379]	-0.028 (0.0493) [0.0481]	0.070 (0.0580) [0.0564]
F-stat ($T_{wmin} = T_{wmax}$)	0.00	0.42	0.04	0.63
Observations	600	600	600	600
<i>Cool (low T_{wmin})</i>				
T_{wmin}	-0.120*** (0.0311) [0.0307]	0.116*** (0.0242) [0.0237]	-0.018 (0.0370) [0.0364]	0.026 (0.0328) [0.0321]
T_{wmax}	0.148*** (0.0457) [0.0469]	-0.112*** (0.0319) [0.0317]	0.003 (0.0468) [0.0462]	-0.044 (0.0408) [0.0403]
F-stat ($T_{wmin} = T_{wmax}$)	14.03	18.55	0.07	0.96
Observations	587	587	587	587
<i>Wet (high baseline rain)</i>				
T_{wmin}	-0.053 (0.0377) [0.0365]	0.040 (0.0302) [0.0296]	0.010 (0.0401) [0.0391]	0.005 (0.0271) [0.0267]
T_{wmax}	0.050 (0.0573) [0.0553]	-0.019 (0.0453) [0.0444]	-0.059 (0.0606) [0.0592]	0.025 (0.0435) [0.0429]
F-stat ($T_{wmin} = T_{wmax}$)	1.24	0.65	0.49	0.09
Observations	591	591	591	591
<i>Dry (low rain)</i>				
T_{wmin}	-0.048* (0.0288) [0.0286]	0.089*** (0.0201) [0.0199]	-0.062* (0.0316) [0.0310]	0.025 (0.0330) [0.0323]
T_{wmax}	0.060 (0.0479) [0.0484]	-0.100*** (0.0316) [0.0310]	0.074* (0.0408) [0.0395]	-0.038 (0.0482) [0.0470]
F-stat ($T_{wmin} = T_{wmax}$)	2.16	14.80	3.87	0.65
Observations	596	596	596	596
<i>North (high latitude)</i>				
T_{wmin}	-0.143*** (0.0314) [0.0312]	0.146*** (0.0252) [0.0247]	-0.014 (0.0360) [0.0355]	0.021 (0.0324) [0.0318]
T_{wmax}	0.154*** (0.0478) [0.0490]	-0.137*** (0.0337) [0.0333]	0.006 (0.0477) [0.0473]	-0.035 (0.0418) [0.0412]
F-stat ($T_{wmin} = T_{wmax}$)	15.44	24.96	0.06	0.61
Observations	586	586	586	586
<i>South (low latitude)</i>				
T_{wmin}	-0.072** (0.0294) [0.0286]	0.028 (0.0304) [0.0296]	0.023 (0.0279) [0.0273]	0.016 (0.0352) [0.0344]
T_{wmax}	0.056 (0.0460) [0.0448]	-0.023 (0.0424) [0.0413]	-0.070 (0.0431) [0.0422]	0.045 (0.0542) [0.0528]
F-stat ($T_{wmin} = T_{wmax}$)	3.16	0.54	1.97	0.11
Observations	601	601	601	601
Dep. var. mean	0.173	0.180	0.323	0.324

Notes: Blocks restrict the estimation sample by baseline (long-run) wet-bulb temperature, baseline rainfall, latitude, and, where available, baseline irrigation or 1961 urban status. Decadal panel (1981–2011) with unit and decade fixed effects and a state $\times$ decade linear trend. Temperatures are wet-bulb unless noted, constructed as a moving average of the preceding decade's growing seasons (June–February), as inverse-distance-squared weighted averages of surrounding IMD grid points (see Appendix C.3). All specifications control for total precipitation over the same window unless noted. Standard errors in parentheses, clustered at the district level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

TABLE A40. ICRI SAT Districts, Long Difference: Heterogeneity

	Cultivated Area (1)	Yield Index (2)	Field Wage (3)	Price Index (4)
<i>All units</i>				
$\Delta T_{\min}$	0.289 (0.2017) [0.1954]	0.357** (0.1668) [0.1601]	0.144 (0.1904) [0.1849]	-0.118 (0.1013) [0.0981]
$\Delta T_{\max}$	-1.715*** (0.6231) [0.6034]	-0.626 (0.5064) [0.4897]	-0.675** (0.3403) [0.3204]	0.138 (0.2002) [0.1943]
F-stat ($\Delta T_{\min} = \Delta T_{\max}$)	6.15	2.23	2.57	0.76
Observations	311	303	259	294
<i>Hot (high baseline $T_{\min}$)</i>				
$\Delta T_{\min}$	0.265 (0.4278) [0.4049]	0.263 (0.2905) [0.2683]	0.163 (0.2696) [0.2493]	-0.218 (0.1508) [0.1560]
$\Delta T_{\max}$	-2.030* (1.1102) [1.0518]	-0.854 (0.8608) [0.8137]	-0.786* (0.4596) [0.4265]	0.190 (0.2683) [0.2657]
F-stat ($\Delta T_{\min} = \Delta T_{\max}$)	2.34	0.98	1.81	1.01
Observations	155	153	145	145
<i>Cool (low $T_{\min}$)</i>				
$\Delta T_{\min}$	0.176 (0.1596) [0.1528]	0.362*** (0.1021) [0.0997]	0.085 (0.2851) [0.2733]	-0.086 (0.1519) [0.1414]
$\Delta T_{\max}$	-0.971** (0.4163) [0.3897]	-0.087 (0.2468) [0.2397]	-0.563 (0.6344) [0.5941]	0.289 (0.3620) [0.3458]
F-stat ($\Delta T_{\min} = \Delta T_{\max}$)	4.32	2.03	0.54	0.55
Observations	151	144	109	143
<i>Wet (high baseline rain)</i>				
$\Delta T_{\min}$	-0.377 (0.3728) [0.3391]	-0.228 (0.2520) [0.2286]	0.444 (0.3137) [0.2795]	-0.275 (0.2942) [0.2701]
$\Delta T_{\max}$	-0.126 (0.7392) [0.6796]	0.862* (0.4735) [0.4322]	-0.804 (0.5822) [0.4934]	0.255 (0.4440) [0.4226]
F-stat ($\Delta T_{\min} = \Delta T_{\max}$)	0.05	2.31	2.02	0.55
Observations	155	148	133	142
<i>Dry (low rain)</i>				
$\Delta T_{\min}$	0.490 (0.3113) [0.2955]	0.652** (0.2557) [0.2407]	0.132 (0.2621) [0.2476]	-0.082 (0.1250) [0.1184]
$\Delta T_{\max}$	-2.339** (1.1780) [1.1185]	-1.475 (0.9941) [0.9430]	-0.863* (0.4928) [0.4573]	0.099 (0.2645) [0.2500]
F-stat ($\Delta T_{\min} = \Delta T_{\max}$)	3.73	2.98	1.88	0.23
Observations	155	154	125	151
<i>North (high latitude)</i>				
$\Delta T_{\min}$	0.325** (0.1599) [0.1454]	0.264** (0.1397) [0.1271]	0.201 (0.2655) [0.2479]	-0.102 (0.1490) [0.1389]
$\Delta T_{\max}$	-1.218*** (0.3808) [0.3609]	-0.103 (0.2881) [0.2774]	-0.506 (0.6190) [0.5623]	0.145 (0.3395) [0.3262]
F-stat ($\Delta T_{\min} = \Delta T_{\max}$)	9.00	0.84	0.71	0.27
Observations	154	146	107	142
<i>South (low latitude)</i>				
$\Delta T_{\min}$	-0.226 (0.4304) [0.4073]	0.208 (0.2678) [0.2480]	-0.048 (0.2842) [0.2544]	-0.433*** (0.1593) [0.1562]
$\Delta T_{\max}$	-1.363 (0.9966) [0.9483]	-0.727 (0.7604) [0.7202]	-0.529 (0.4505) [0.4157]	0.442* (0.2639) [0.2595]
F-stat ($\Delta T_{\min} = \Delta T_{\max}$)	0.70	0.90	0.45	4.54
Observations	155	155	150	150
<i>High baseline irrigation</i>				
$\Delta T_{\min}$	0.127 (0.1932) [0.1816]	0.351* (0.1995) [0.1880]	0.404 (0.2570) [0.2396]	0.015 (0.1358) [0.1227]
$\Delta T_{\max}$	-1.320*** (0.4669) [0.4410]	-0.295 (0.3748) [0.3531]	-0.853* (0.4758) [0.4449]	0.011 (0.2603) [0.2428]
F-stat ($\Delta T_{\min} = \Delta T_{\max}$)	5.37	1.43	3.24	0.00
Observations	165	158	135	158
<i>Low baseline irrigation</i>				
$\Delta T_{\min}$	0.163 (0.2545) [0.2375]	0.142 (0.1143) [0.0948]	-0.046 (0.3384) [0.3196]	-0.375** (0.1646) [0.1565]
$\Delta T_{\max}$	-1.063* (0.5636) [0.5275]	-0.010 (0.2263) [0.1928]	-0.581 (0.6780) [0.6113]	0.504 (0.3325) [0.3134]
F-stat ($\Delta T_{\min} = \Delta T_{\max}$)	2.34	0.21	0.29	3.21
Observations	123	123	104	116
Dep. var. mean	-0.076	0.368	1.919	1.364

Notes: Blocks restrict the estimation sample by baseline (long-run) wet-bulb temperature, baseline rainfall, latitude, and, where available, baseline irrigation or 1961 urban status. Long-differenced cross-section: outcomes and climate averaged over 1981–91 (baseline) and 2001–11 (endline), then differenced. All specifications include state fixed effects. Temperatures are wet-bulb unless noted, constructed as a moving average of the preceding decade’s growing seasons (June–February), as inverse-distance-squared weighted averages of surrounding IMD grid points (see Appendix C.3). All specifications control for total precipitation over the same window unless noted. Standard errors in parentheses, clustered at the district level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

TABLE A41. ICRISAT Districts, Panel: Heterogeneity

	Cultivated Area (1)	Yield Index (2)	Field Wage (3)	Price Index (4)
<i>All units</i>				
T_{wmin}	0.090 (0.0781) [0.0778]	0.153** (0.0773) [0.0770]	0.137 (0.1243) [0.1204]	-0.237*** (0.0598) [0.0598]
T_{wmax}	-0.579*** (0.1534) [0.1529]	-0.194 (0.1280) [0.1270]	-0.328* (0.1928) [0.1854]	0.366*** (0.0936) [0.0946]
F-stat ($T_{wmin} = T_{wmax}$)	9.01	3.07	2.28	16.59
Observations	1,243	1,211	1,115	1,191
<i>Hot (high baseline T_{wmin})</i>				
T_{wmin}	0.118 (0.1444) [0.1436]	0.160 (0.1196) [0.1164]	0.170 (0.1438) [0.1484]	-0.473*** (0.0823) [0.0865]
T_{wmax}	-0.540** (0.2895) [0.2869]	-0.373 (0.2269) [0.2195]	-0.478** (0.2202) [0.2231]	0.731*** (0.1216) [0.1282]
F-stat ($T_{wmin} = T_{wmax}$)	2.38	2.48	3.42	38.65
Observations	619	615	592	598
<i>Cool (low T_{wmin})</i>				
T_{wmin}	0.091 (0.1010) [0.0998]	0.264*** (0.0874) [0.0871]	0.062 (0.2116) [0.1913]	-0.089 (0.0841) [0.0817]
T_{wmax}	-0.667*** (0.1599) [0.1594]	-0.057 (0.1277) [0.1279]	-0.397 (0.3437) [0.3029]	0.120 (0.1442) [0.1439]
F-stat ($T_{wmin} = T_{wmax}$)	9.43	2.49	0.73	0.92
Observations	624	596	523	593
<i>Wet (high baseline rain)</i>				
T_{wmin}	-0.273** (0.1223) [0.1176]	-0.055 (0.1206) [0.1157]	0.176 (0.1576) [0.1508]	-0.321* (0.1760) [0.1737]
T_{wmax}	0.077 (0.1884) [0.1829]	0.175 (0.1768) [0.1690]	-0.170 (0.2455) [0.2393]	0.454** (0.2279) [0.2278]
F-stat ($T_{wmin} = T_{wmax}$)	1.33	0.63	0.77	3.75
Observations	624	596	559	583
<i>Dry (low rain)</i>				
T_{wmin}	0.250** (0.1096) [0.1083]	0.267** (0.1035) [0.1028]	0.156 (0.1797) [0.1724]	-0.182*** (0.0685) [0.0663]
T_{wmax}	-0.847*** (0.2676) [0.2560]	-0.441** (0.2092) [0.2022]	-0.382 (0.2964) [0.2757]	0.324*** (0.1212) [0.1207]
F-stat ($T_{wmin} = T_{wmax}$)	9.29	5.76	1.40	7.80
Observations	619	615	556	608
<i>North (high latitude)</i>				
T_{wmin}	0.232*** (0.0889) [0.0894]	0.196** (0.0926) [0.0926]	-0.016 (0.1963) [0.1776]	-0.154* (0.0886) [0.0871]
T_{wmax}	-0.742*** (0.1426) [0.1426]	-0.096 (0.1369) [0.1358]	-0.088 (0.3031) [0.2700]	0.126 (0.1391) [0.1385]
F-stat ($T_{wmin} = T_{wmax}$)	19.06	1.74	0.02	1.62
Observations	624	592	508	583
<i>South (low latitude)</i>				
T_{wmin}	-0.032 (0.1753) [0.1749]	0.201 (0.1301) [0.1276]	0.269 (0.1705) [0.1741]	-0.367*** (0.0859) [0.0874]
T_{wmax}	-0.394 (0.3317) [0.3392]	-0.385 (0.2421) [0.2360]	-0.653*** (0.2285) [0.2329]	0.704*** (0.1210) [0.1237]
F-stat ($T_{wmin} = T_{wmax}$)	0.54	2.62	5.86	28.96
Observations	619	619	607	608
<i>High baseline irrigation</i>				
T_{wmin}	0.126 (0.0941) [0.0929]	0.240* (0.1269) [0.1249]	0.382** (0.1830) [0.1747]	-0.177** (0.0810) [0.0797]
T_{wmax}	-0.725*** (0.1720) [0.1709]	-0.281 (0.1810) [0.1785]	-0.762*** (0.2799) [0.2665]	0.348** (0.1428) [0.1419]
F-stat ($T_{wmin} = T_{wmax}$)	11.41	3.11	6.55	5.82
Observations	664	636	586	633
<i>Low baseline irrigation</i>				
T_{wmin}	0.033 (0.1083) [0.1066]	-0.013 (0.0725) [0.0707]	-0.170 (0.1734) [0.1627]	-0.366*** (0.0781) [0.0767]
T_{wmax}	-0.379** (0.1611) [0.1589]	0.091 (0.1123) [0.1094]	0.177 (0.3011) [0.2796]	0.545*** (0.1231) [0.1212]
F-stat ($T_{wmin} = T_{wmax}$)	2.48	0.34	0.56	22.29
Observations	499	499	453	482
Dep. var. mean	5.451	7.142	2.856	5.660

Notes: Blocks restrict the estimation sample by baseline (long-run) wet-bulb temperature, baseline rainfall, latitude, and, where available, baseline irrigation or 1961 urban status. Decadal panel (1981–2011) with unit and decade fixed effects and a state $\times$ decade linear trend. Temperatures are wet-bulb unless noted, constructed as a moving average of the preceding decade's growing seasons (June–February), as inverse-distance-squared weighted averages of surrounding IMD grid points (see Appendix C.3). All specifications control for total precipitation over the same window unless noted. Standard errors in parentheses, clustered at the district level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

TABLE A42. NSS Urban Households, Panel: Heterogeneity

	Agriculture (1)	Constr. & Mining (2)	Manufacturing (3)	Services (4)
<i>All units</i>				
T_{wmin}	0.160*** (0.0614) [0.0600]	0.060 (0.0408) [0.0395]	-0.064 (0.0485) [0.0467]	-0.155** (0.0725) [0.0698]
T_{wmax}	-0.196* (0.1092) [0.1052]	-0.105 (0.0799) [0.0770]	0.091 (0.0880) [0.0846]	0.210 (0.1321) [0.1281]
F-stat ($T_{wmin} = T_{wmax}$)	4.63	1.99	1.41	3.42
Observations	1,163	1,163	1,163	1,163
<i>Hot (high baseline T_{wmin})</i>				
T_{wmin}	0.277*** (0.0979) [0.0962]	0.030 (0.0573) [0.0556]	-0.115 (0.0885) [0.0869]	-0.192 (0.1250) [0.1199]
T_{wmax}	-0.336** (0.1601) [0.1532]	-0.024 (0.1094) [0.1046]	0.239 (0.1491) [0.1440]	0.122 (0.1990) [0.1898]
F-stat ($T_{wmin} = T_{wmax}$)	6.01	0.11	2.36	0.99
Observations	594	594	594	594
<i>Cool (low T_{wmin})</i>				
T_{wmin}	0.090 (0.0818) [0.0781]	0.080 (0.0602) [0.0582]	-0.012 (0.0590) [0.0548]	-0.157* (0.0940) [0.0902]
T_{wmax}	-0.084 (0.1650) [0.1577]	-0.227* (0.1287) [0.1238]	-0.111 (0.1240) [0.1161]	0.422** (0.2062) [0.1986]
F-stat ($T_{wmin} = T_{wmax}$)	0.54	2.93	0.33	4.16
Observations	569	569	569	569
<i>Wet (high baseline rain)</i>				
T_{wmin}	0.164* (0.0936) [0.0900]	0.048 (0.0573) [0.0525]	-0.065 (0.0788) [0.0752]	-0.146 (0.1192) [0.1105]
T_{wmax}	-0.229 (0.1680) [0.1564]	-0.087 (0.1190) [0.1086]	0.194 (0.1423) [0.1340]	0.123 (0.2225) [0.2078]
F-stat ($T_{wmin} = T_{wmax}$)	2.38	0.61	1.52	0.65
Observations	575	575	575	575
<i>Dry (low rain)</i>				
T_{wmin}	0.177** (0.0888) [0.0863]	0.115* (0.0616) [0.0605]	-0.097 (0.0669) [0.0637]	-0.195* (0.0992) [0.0963]
T_{wmax}	-0.247 (0.1582) [0.1543]	-0.166 (0.1215) [0.1192]	0.049 (0.1232) [0.1181]	0.364* (0.1852) [0.1818]
F-stat ($T_{wmin} = T_{wmax}$)	3.23	2.56	0.64	4.20
Observations	588	588	588	588
<i>North (high latitude)</i>				
T_{wmin}	0.146* (0.0812) [0.0799]	0.068 (0.0541) [0.0526]	-0.007 (0.0579) [0.0546]	-0.207** (0.0917) [0.0891]
T_{wmax}	-0.131 (0.1608) [0.1562]	-0.163 (0.1213) [0.1182]	-0.061 (0.1140) [0.1080]	0.355* (0.1978) [0.1931]
F-stat ($T_{wmin} = T_{wmax}$)	1.46	1.88	0.11	4.21
Observations	570	570	570	570
<i>South (low latitude)</i>				
T_{wmin}	0.335*** (0.1079) [0.1021]	-0.009 (0.0726) [0.0721]	-0.271*** (0.0937) [0.0899]	-0.055 (0.1248) [0.1194]
T_{wmax}	-0.410** (0.1637) [0.1533]	0.006 (0.1184) [0.1151]	0.338** (0.1485) [0.1423]	0.066 (0.1908) [0.1826]
F-stat ($T_{wmin} = T_{wmax}$)	7.93	0.01	6.78	0.15
Observations	593	593	593	593
Dep. var. mean	0.112	0.094	0.170	0.624

Notes: Blocks restrict the estimation sample by baseline (long-run) wet-bulb temperature, baseline rainfall, latitude, and, where available, baseline irrigation or 1961 urban status. Decadal panel (1981–2011) with unit and decade fixed effects and a state $\times$ decade linear trend. Temperatures are wet-bulb unless noted, constructed as a moving average of the preceding decade's growing seasons (June–February), as inverse-distance-squared weighted averages of surrounding IMD grid points (see Appendix C.3). All specifications control for total precipitation over the same window unless noted. Standard errors in parentheses, clustered at the district level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

TABLE A43. Census Towns, Long Difference: Heterogeneity

	Seasonal (1)	Ag. Labour (2)	Cultivation (3)	Non-Agriculture (4)
<i>All units</i>				
ΔT_{wmin}	-0.042** (0.0209) [0.0231]	0.003 (0.0225) [0.0242]	-0.048** (0.0220) [0.0237]	0.087*** (0.0308) [0.0331]
ΔT_{wmax}	0.051 (0.0390) [0.0432]	0.119*** (0.0419) [0.0463]	0.079** (0.0398) [0.0414]	-0.249*** (0.0612) [0.0648]
F-stat ($\Delta T_{wmin} = \Delta T_{wmax}$)	2.63	3.47	4.50	14.29
Observations	2,969	2,969	2,969	2,969
<i>Hot (high baseline T_{wmin})</i>				
ΔT_{wmin}	-0.040 (0.0311) [0.0317]	0.030 (0.0389) [0.0395]	0.000 (0.0258) [0.0240]	0.008 (0.0531) [0.0515]
ΔT_{wmax}	0.030 (0.0532) [0.0537]	0.090 (0.0638) [0.0691]	0.020 (0.0379) [0.0370]	-0.141 (0.0862) [0.0889]
F-stat ($\Delta T_{wmin} = \Delta T_{wmax}$)	0.74	0.37	0.11	1.21
Observations	1,485	1,485	1,485	1,485
<i>Cool (low T_{wmin})</i>				
ΔT_{wmin}	-0.044 (0.0275) [0.0308]	-0.006 (0.0285) [0.0304]	-0.071** (0.0313) [0.0338]	0.122*** (0.0400) [0.0427]
ΔT_{wmax}	0.098 (0.0595) [0.0689]	0.129** (0.0621) [0.0663]	0.104 (0.0721) [0.0744]	-0.326*** (0.0980) [0.0991]
F-stat ($\Delta T_{wmin} = \Delta T_{wmax}$)	3.00	2.39	3.07	11.33
Observations	1,484	1,484	1,484	1,484
<i>Wet (high baseline rain)</i>				
ΔT_{wmin}	-0.039 (0.0352) [0.0351]	-0.077** (0.0372) [0.0376]	-0.084*** (0.0313) [0.0363]	0.203*** (0.0481) [0.0535]
ΔT_{wmax}	-0.004 (0.0690) [0.0706]	0.339*** (0.0731) [0.0755]	0.173*** (0.0557) [0.0666]	-0.519*** (0.1033) [0.1197]
F-stat ($\Delta T_{wmin} = \Delta T_{wmax}$)	0.12	14.90	9.22	24.20
Observations	1,485	1,485	1,485	1,485
<i>Dry (low rain)</i>				
ΔT_{wmin}	-0.034 (0.0259) [0.0286]	0.030 (0.0272) [0.0282]	-0.022 (0.0260) [0.0264]	0.022 (0.0387) [0.0379]
ΔT_{wmax}	0.084* (0.0505) [0.0555]	-0.018 (0.0514) [0.0540]	0.011 (0.0542) [0.0530]	-0.067 (0.0777) [0.0737]
F-stat ($\Delta T_{wmin} = \Delta T_{wmax}$)	2.58	0.39	0.18	0.62
Observations	1,482	1,482	1,482	1,482
<i>North (high latitude)</i>				
ΔT_{wmin}	-0.039 (0.0275) [0.0311]	-0.007 (0.0278) [0.0294]	-0.072** (0.0299) [0.0324]	0.118*** (0.0392) [0.0422]
ΔT_{wmax}	0.021 (0.0645) [0.0743]	0.100 (0.0620) [0.0653]	0.073 (0.0693) [0.0723]	-0.193* (0.0983) [0.1017]
F-stat ($\Delta T_{wmin} = \Delta T_{wmax}$)	0.48	1.53	2.27	5.46
Observations	1,485	1,485	1,485	1,485
<i>South (low latitude)</i>				
ΔT_{wmin}	-0.060* (0.0347) [0.0353]	0.073* (0.0435) [0.0431]	0.045* (0.0243) [0.0247]	-0.062 (0.0568) [0.0538]
ΔT_{wmax}	0.091 (0.0565) [0.0566]	0.043 (0.0651) [0.0703]	-0.014 (0.0354) [0.0357]	-0.117 (0.0865) [0.0864]
F-stat ($\Delta T_{wmin} = \Delta T_{wmax}$)	2.86	0.08	1.04	0.16
Observations	1,483	1,483	1,483	1,483
Dep. var. mean	0.102	-0.053	-0.058	0.010

Notes: Blocks restrict the estimation sample by baseline (long-run) wet-bulb temperature, baseline rainfall, latitude, and, where available, baseline irrigation or 1961 urban status. Long-differenced cross-section: outcomes and climate averaged over 1981–91 (baseline) and 2001–11 (endline), then differenced. All specifications include state fixed effects. Temperatures are wet-bulb unless noted, constructed as a moving average of the preceding decade’s growing seasons (June–February), as inverse-distance-squared weighted averages of surrounding IMD grid points (see Appendix C.3). All specifications control for total precipitation over the same window unless noted. Standard errors in parentheses, clustered at the district level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

TABLE A44. Census Towns, Panel: Heterogeneity

	Seasonal (1)	Ag. Labour (2)	Cultivation (3)	Non-Agriculture (4)
<i>All units</i>				
$T_{\min}$	-0.023** (0.0113) [0.0106]	0.039** (0.0118) [0.0100]	-0.017* (0.0094) [0.0084]	0.008 (0.0160) [0.0148]
$T_{\max}$	0.046** (0.0184) [0.0177]	-0.025 (0.0190) [0.0165]	0.016 (0.0133) [0.0125]	-0.037 (0.0258) [0.0237]
F-stat ($T_{\min} = T_{\max}$)	5.91	3.37	2.15	1.29
Observations	16,215	16,215	16,215	16,215
<i>Hot (high baseline $T_{\min}$)</i>				
$T_{\min}$	-0.053** (0.0258) [0.0206]	0.049** (0.0219) [0.0189]	-0.010 (0.0104) [0.0091]	0.004 (0.0312) [0.0268]
$T_{\max}$	0.047 (0.0367) [0.0285]	-0.096 (0.0344) [0.0285]	0.044*** (0.0143) [0.0129]	-0.022 (0.0496) [0.0394]
F-stat ($T_{\min} = T_{\max}$)	2.70	3.79	5.18	0.11
Observations	6,831	6,831	6,831	6,831
<i>Cool (low $T_{\min}$)</i>				
$T_{\min}$	-0.019 (0.0125) [0.0123]	0.027** (0.0129) [0.0108]	-0.015 (0.0123) [0.0111]	0.009 (0.0175) [0.0166]
$T_{\max}$	0.049** (0.0207) [0.0215]	-0.023 (0.0206) [0.0182]	-0.001 (0.0172) [0.0166]	-0.032 (0.0278) [0.0266]
F-stat ($T_{\min} = T_{\max}$)	4.59	2.35	0.23	0.86
Observations	9,384	9,384	9,384	9,384
<i>Wet (high baseline rain)</i>				
$T_{\min}$	-0.003 (0.0172) [0.0157]	0.021 (0.0207) [0.0177]	-0.022 (0.0139) [0.0118]	-0.001 (0.0297) [0.0267]
$T_{\max}$	0.004 (0.0282) [0.0260]	-0.022 (0.0353) [0.0302]	0.037* (0.0201) [0.0174]	-0.012 (0.0498) [0.0445]
F-stat ($T_{\min} = T_{\max}$)	0.03	0.62	3.11	0.02
Observations	7,439	7,439	7,439	7,439
<i>Dry (low rain)</i>				
$T_{\min}$	-0.023 (0.0146) [0.0133]	0.036** (0.0143) [0.0120]	-0.011 (0.0119) [0.0109]	-0.001 (0.0189) [0.0175]
$T_{\max}$	0.064*** (0.0245) [0.0230]	-0.033 (0.0216) [0.0190]	-0.003 (0.0171) [0.0169]	-0.029 (0.0303) [0.0281]
F-stat ($T_{\min} = T_{\max}$)	5.48	3.87	0.08	0.38
Observations	8,776	8,776	8,776	8,776
<i>North (high latitude)</i>				
$T_{\min}$	-0.023 (0.0142) [0.0140]	0.035** (0.0141) [0.0119]	-0.018 (0.0134) [0.0122]	0.009 (0.0197) [0.0188]
$T_{\max}$	0.049** (0.0234) [0.0242]	-0.033* (0.0220) [0.0196]	0.003 (0.0191) [0.0183]	-0.016 (0.0310) [0.0298]
F-stat ($T_{\min} = T_{\max}$)	4.02	4.91	0.44	0.26
Observations	8,896	8,896	8,896	8,896
<i>South (low latitude)</i>				
$T_{\min}$	-0.044** (0.0202) [0.0159]	0.031 (0.0229) [0.0187]	-0.002 (0.0092) [0.0084]	0.006 (0.0291) [0.0242]
$T_{\max}$	0.046 (0.0295) [0.0238]	-0.002 (0.0345) [0.0280]	0.033** (0.0134) [0.0120]	-0.063 (0.0453) [0.0372]
F-stat ($T_{\min} = T_{\max}$)	3.48	0.36	2.53	0.92
Observations	7,319	7,319	7,319	7,319
<i>Urban in 1961</i>				
$T_{\min}$	-0.013 (0.0111) [0.0106]	0.020* (0.0110) [0.0100]	-0.013* (0.0079) [0.0081]	0.004 (0.0160) [0.0156]
$T_{\max}$	0.031* (0.0181) [0.0177]	-0.023 (0.0185) [0.0177]	0.021* (0.0119) [0.0124]	-0.029 (0.0263) [0.0261]
F-stat ($T_{\min} = T_{\max}$)	2.40	2.28	3.13	0.68
Observations	7,689	7,689	7,689	7,689
<i>Rural in 1961</i>				
$T_{\min}$	-0.032* (0.0170) [0.0157]	0.048*** (0.0179) [0.0146]	-0.021 (0.0168) [0.0134]	0.004 (0.0244) [0.0209]
$T_{\max}$	0.062** (0.0281) [0.0262]	-0.040 (0.0280) [0.0231]	0.012 (0.0233) [0.0195]	-0.035 (0.0388) [0.0331]
F-stat ($T_{\min} = T_{\max}$)	4.68	3.88	0.71	0.41
Observations	8,526	8,526	8,526	8,526
Dep. var. mean	0.095	0.094	0.077	0.732

Notes: Blocks restrict the estimation sample by baseline (long-run) wet-bulb temperature, baseline rainfall, latitude, and, where available, baseline irrigation or 1961 urban status. Decadal panel (1981–2011) with unit and decade fixed effects and a state $\times$ decade linear trend. Temperatures are wet-bulb unless noted, constructed as a moving average of the preceding decade's growing seasons (June–February), as inverse-distance-squared weighted averages of surrounding IMD grid points (see Appendix C.3). All specifications control for total precipitation over the same window unless noted. Standard errors in parentheses, clustered at the district level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

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